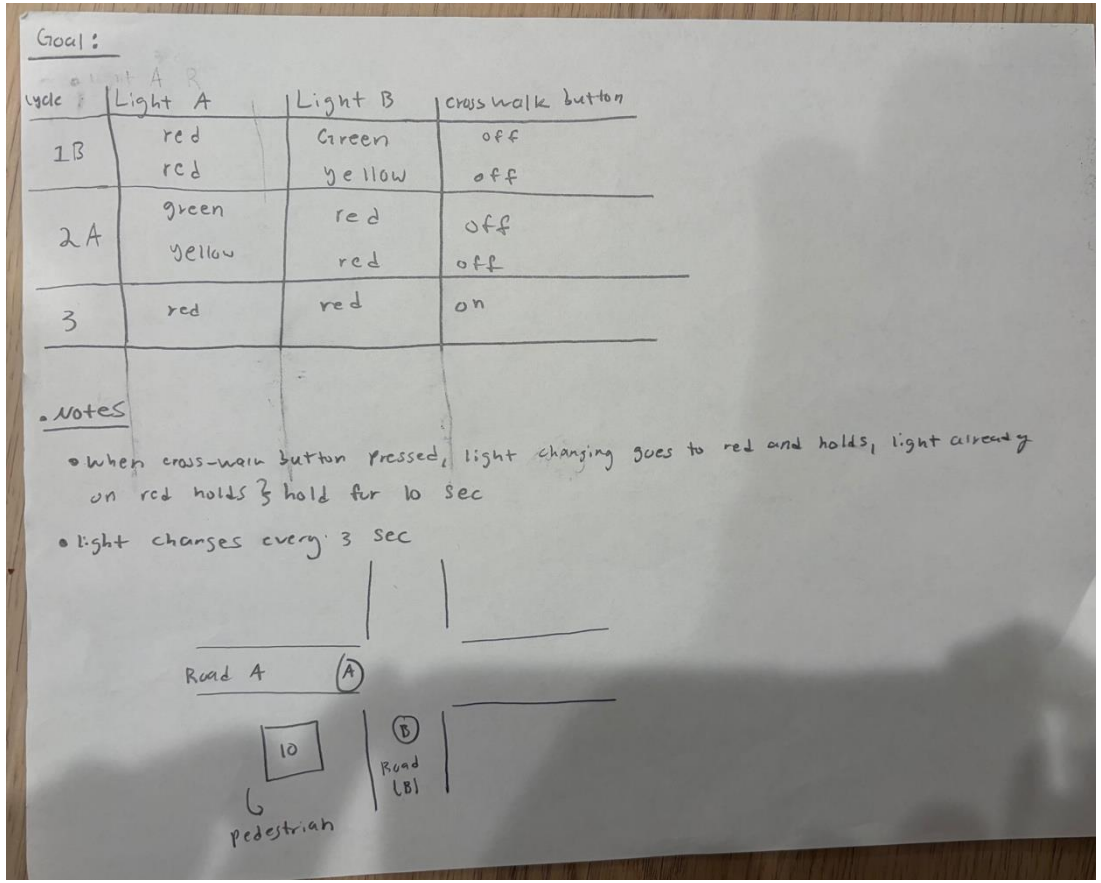
Digital Design Final Project

Grace Dreiling

Trial FSM Drawings

Planning Stage—FSM + Truth Table Trial & Error



Goal:

Cycle	Light A	Light B	crosswalk button
1B	red red	Green yellow	off off
2A	green yellow	red red	off off
3	red	red	on

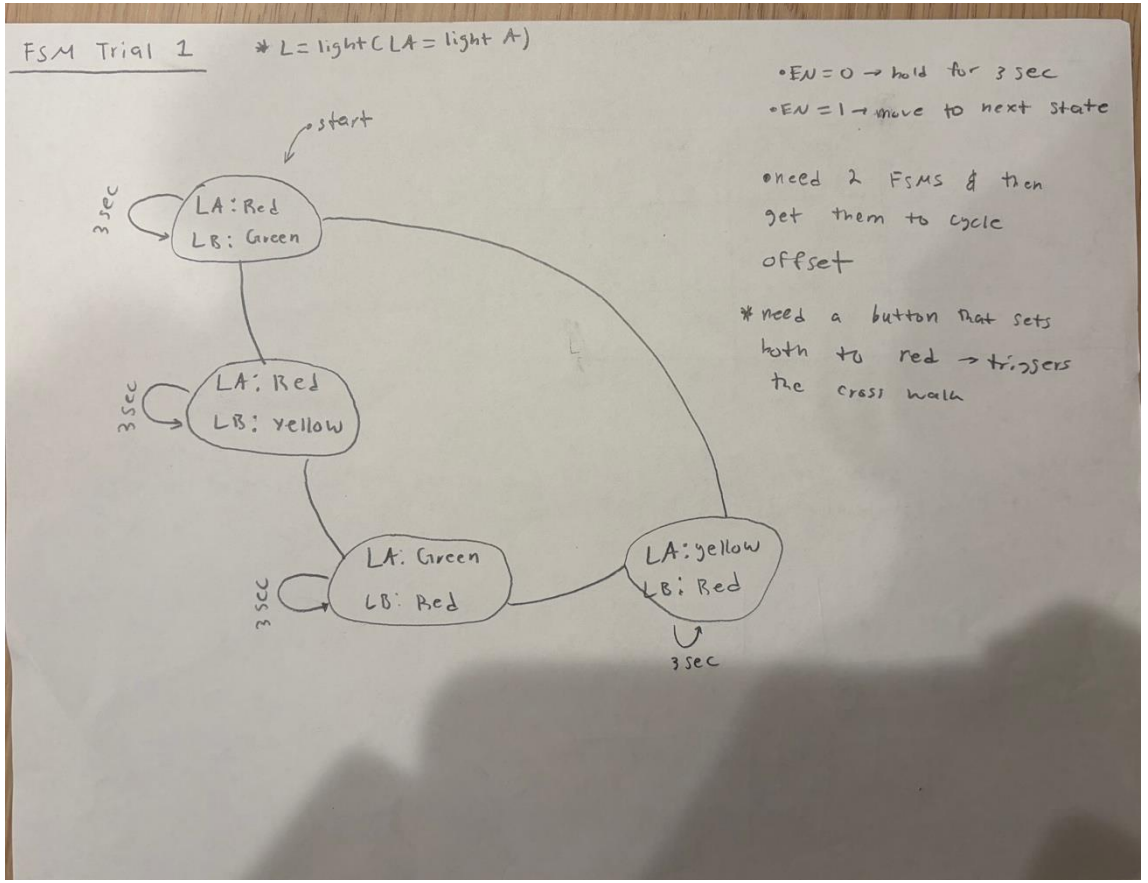
- Notes
  - when cross-walk button pressed, light changing goes to red and holds, light already on red holds & hold for 10 sec
  - light changes every 3 sec

Road A (A)

10

Pedestrian

(B) Road LB



FSM Trial 1

* L = light (LA = light A)

- EN = 0 → hold for 3 sec
- EN = 1 → move to next state

- need 2 FSMs & then get them to cycle offset

* need a button that sets both to red → triggers the cross walk

start

3sec

LA: Red
LB: Green

3sec

LA: Red
LB: yellow

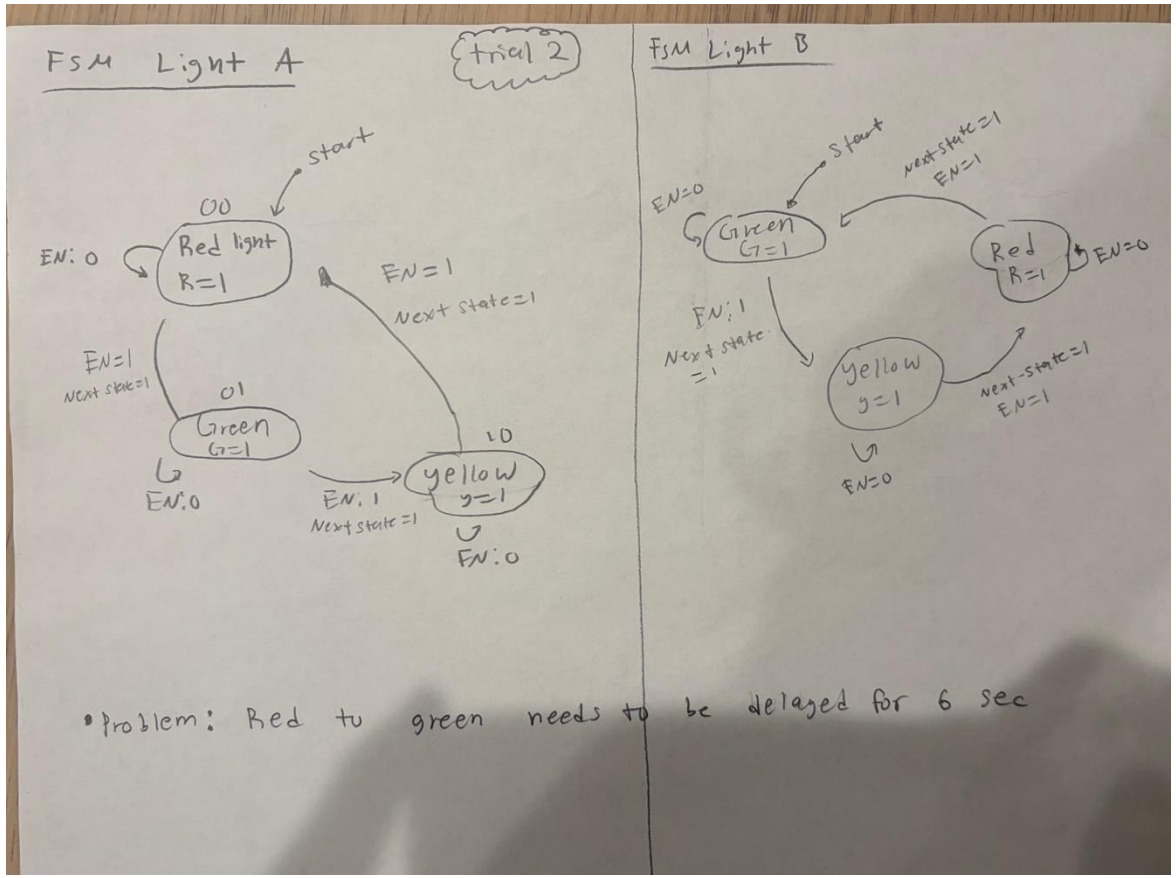
3sec

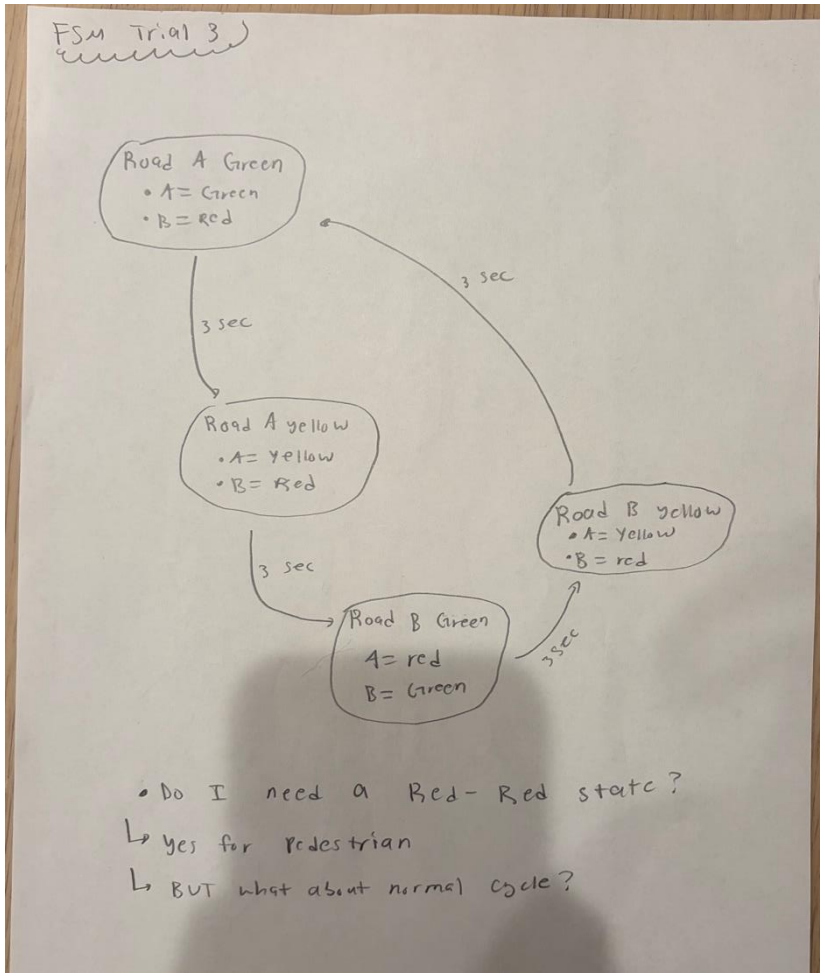
LA: Green
LB: Red

LA: yellow
LB: Red

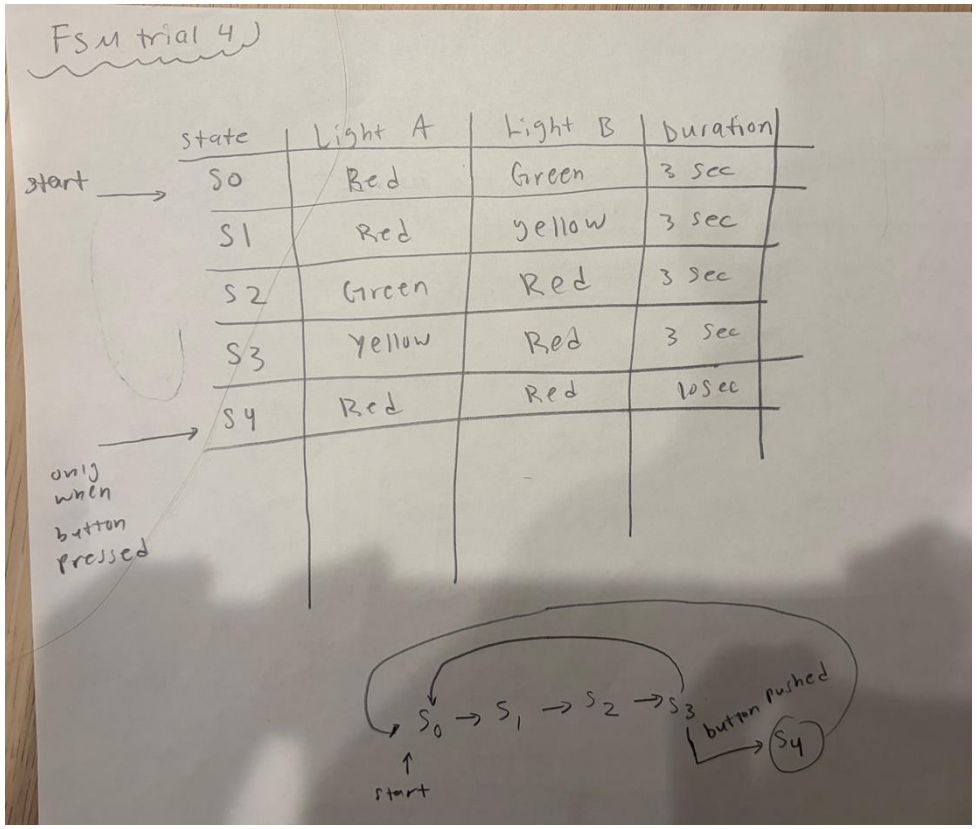
3 sec

Planning Stage—FSM + Truth Table Trial & Error

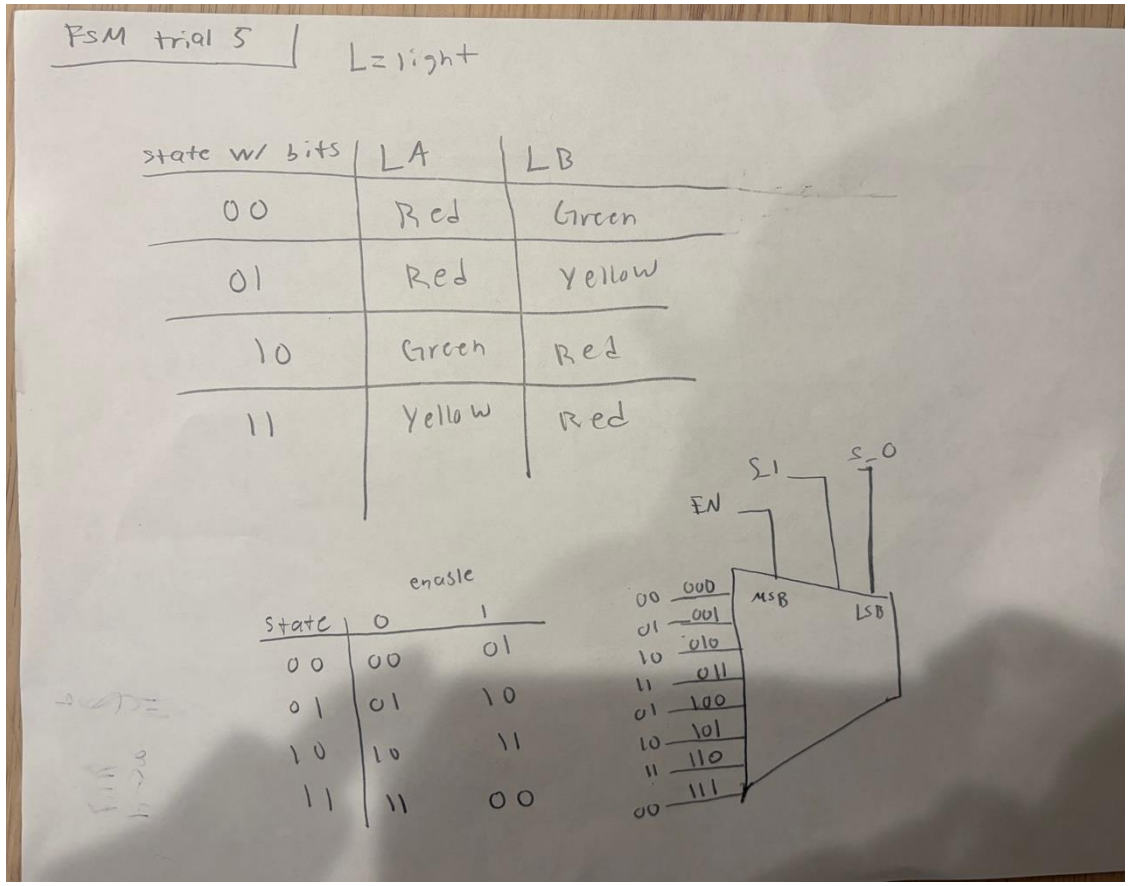




Planning Stage—FSM + Truth Table Trial & Error



state	Light A	Light B	Duration
S0	Red	Green	3 sec
S1	Red	yellow	3 sec
S2	Green	Red	3 sec
S3	yellow	Red	3 sec
S4	Red	Red	10sec



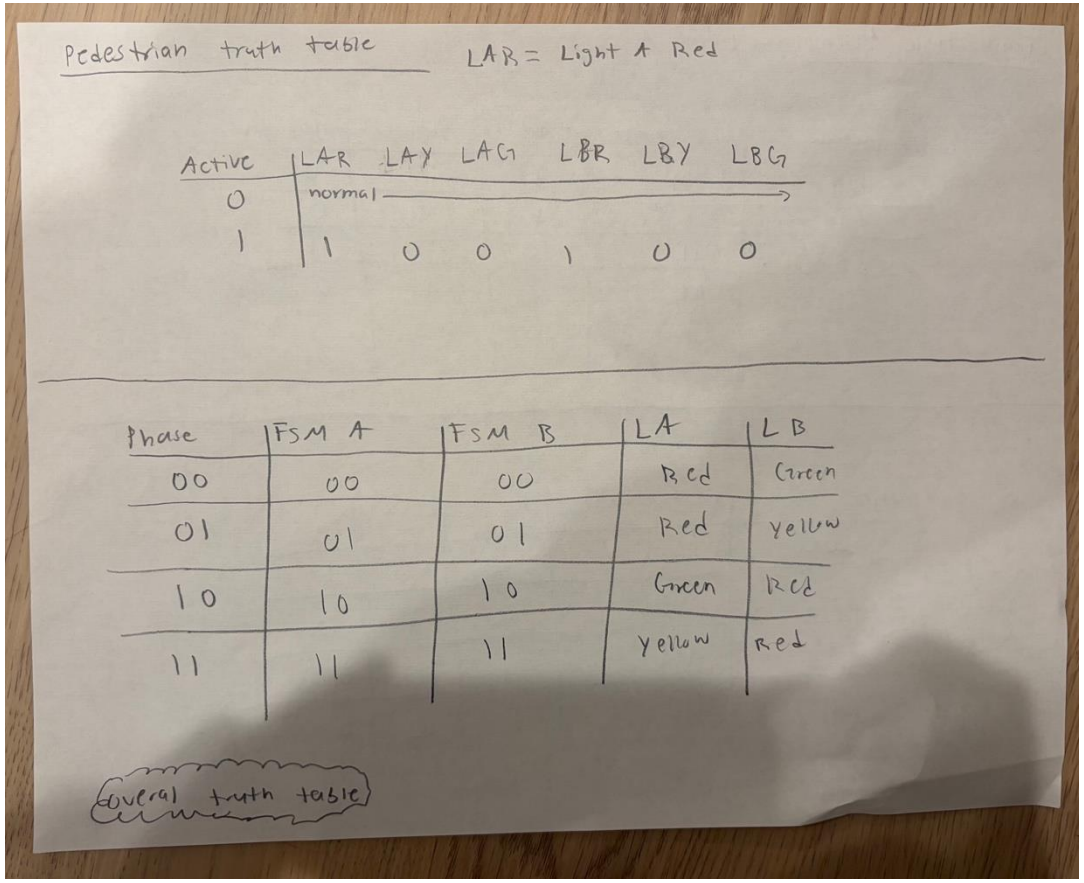
state w/ bits	LA	LB
00	Red	Green
01	Red	Yellow
10	Green	Red
11	Yellow	Red

enasle

State	0	1
00	00	01
01	01	10
10	10	11
11	11	00

00	000
01	001
10	010
11	011
01	100
10	101
11	110
00	111

Planning Stage—FSM + Truth Table Trial & Error



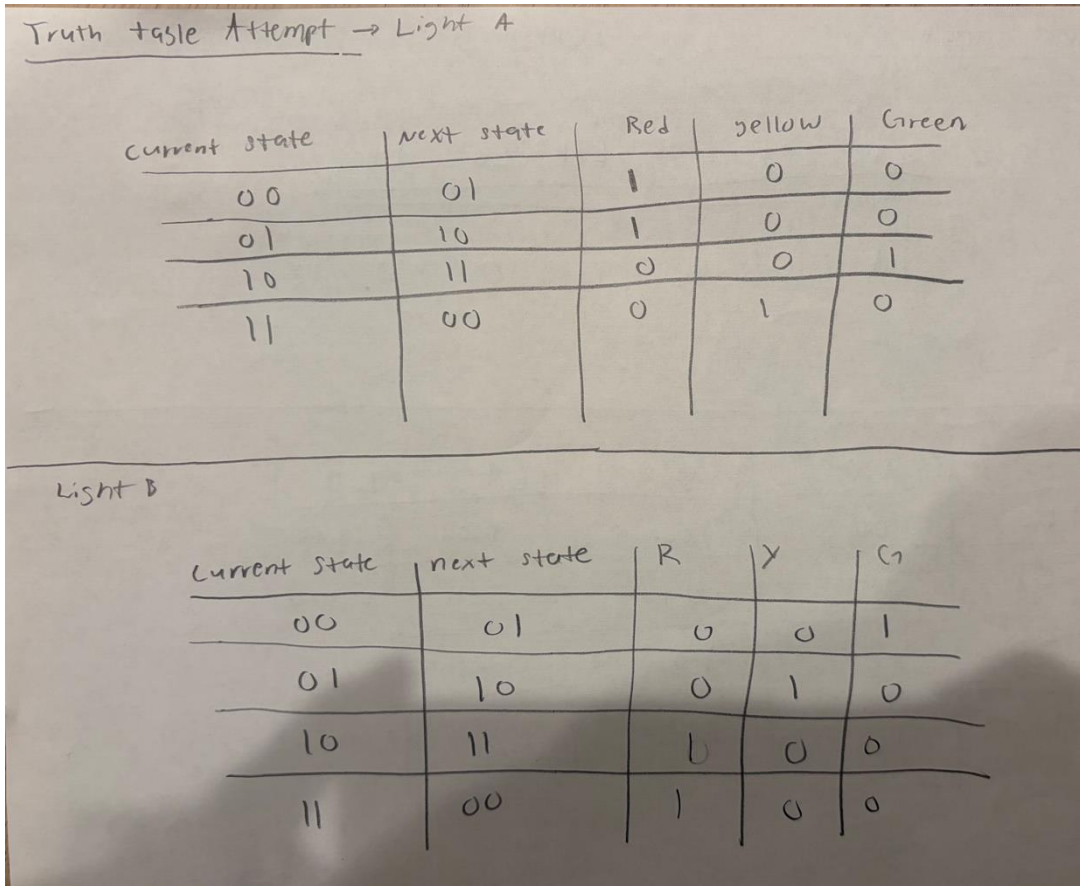
Pedestrian truth table

LAR = Light A Red

Active	LAR	LAY	LAG	LBR	LBY	LBG
0	normal →					
1	1	0	0	1	0	0

Phase	FSM A	FSM B	LA	LB
00	00	00	Red	Green
01	01	01	Red	yellow
10	10	10	Green	Red
11	11	11	yellow	Red

overal truth table



Truth table Attempt → Light A

current state	next state	Red	yellow	Green
00	01	1	0	0
01	10	1	0	0
10	11	0	0	1
11	00	0	1	0

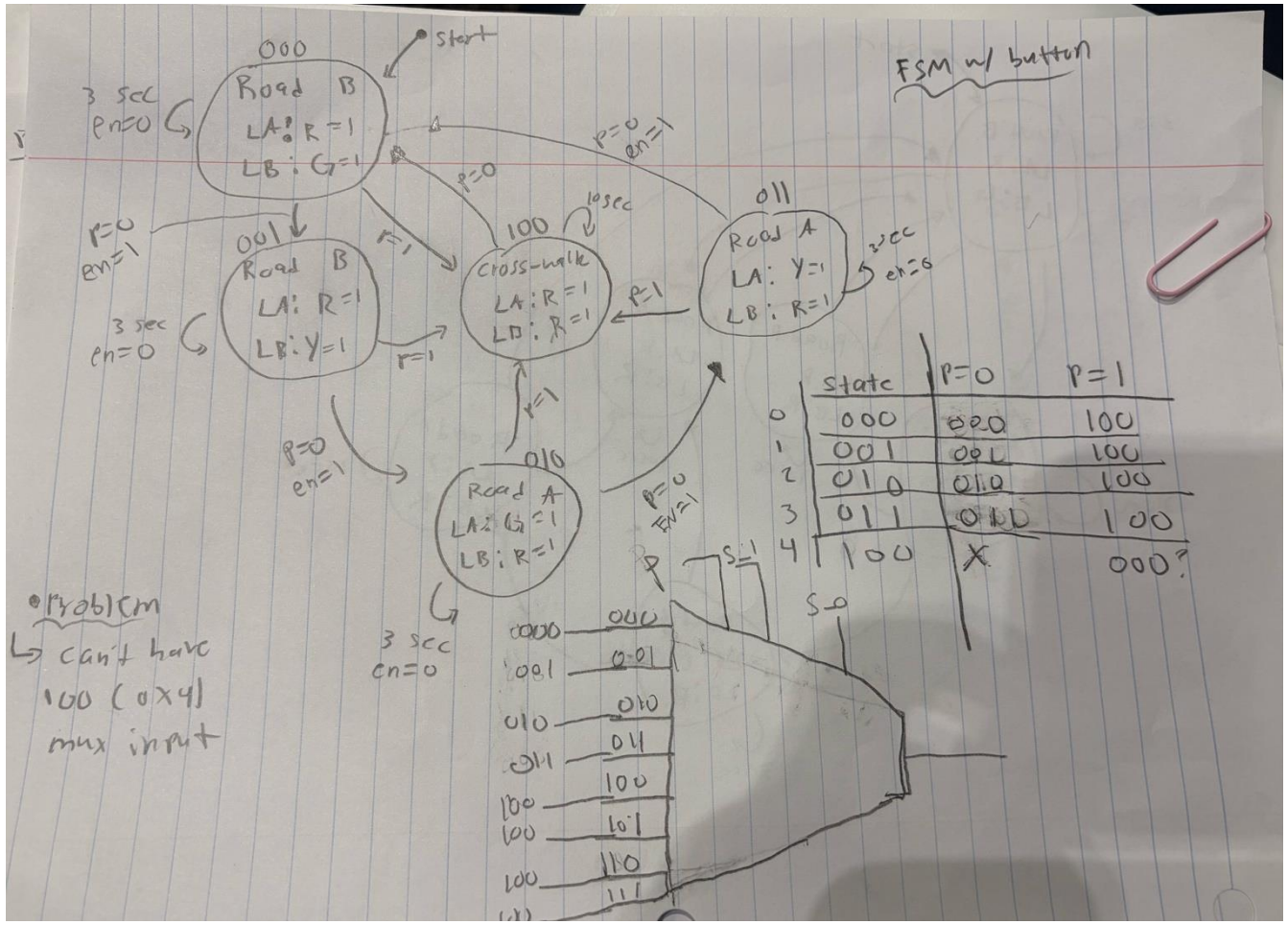
Light B

current state	next state	R	Y	G
00	01	0	0	1
01	10	0	1	0
10	11	0	0	0
11	00	1	0	0

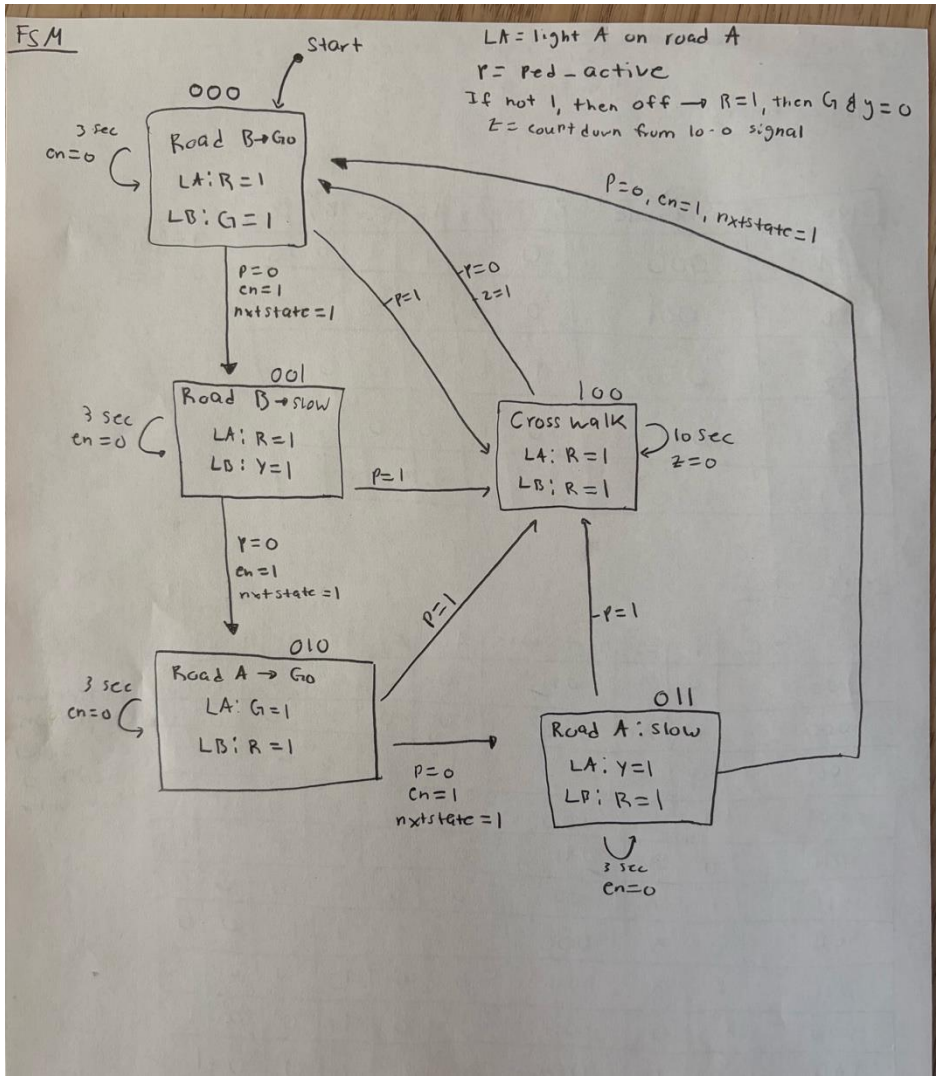
Trying to add in pedestrian button

Problem: if I try to design it this way, the MUX can't handle a 0x4 state for P=1

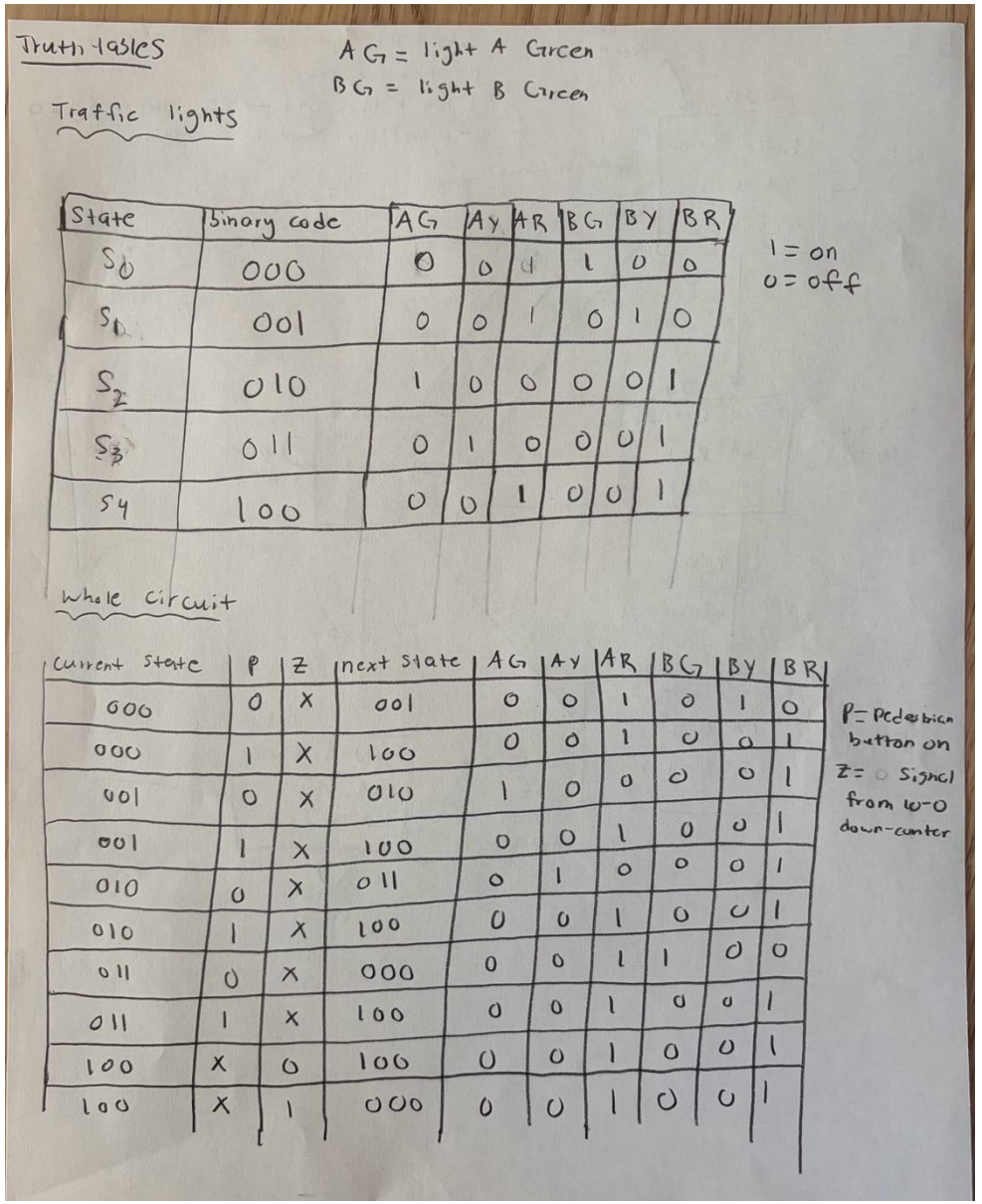
Solution: I need to make sure all software is in 3-bits



Getting Closer to What I Want FSM



Truth Tables



Truth tables

AG = light A Green

BG = light B Green

Traffic lights

State	Binary code	AG	AY	AR	BG	BY	BR
S_0	000	0	0	1	1	0	0
S_1	001	0	0	1	0	1	0
S_2	010	1	0	0	0	0	1
S_3	011	0	1	0	0	0	1
S_4	100	0	0	1	0	0	1

1 = on
0 = off

Whole circuit

current state	P	Z	next state	AG	AY	AR	BG	BY	BR
000	0	X	001	0	0	1	0	1	0
000	1	X	100	0	0	1	0	0	1
001	0	X	010	1	0	0	0	0	1
001	1	X	100	0	0	1	0	0	1
010	0	X	011	0	1	0	0	0	1
010	1	X	100	0	0	1	0	0	1
011	0	X	000	0	0	1	1	0	0
011	1	X	100	0	0	1	0	0	1
100	X	0	100	0	0	1	0	0	1
100	X	1	000	0	0	1	0	0	1

P = Pedestrian button on

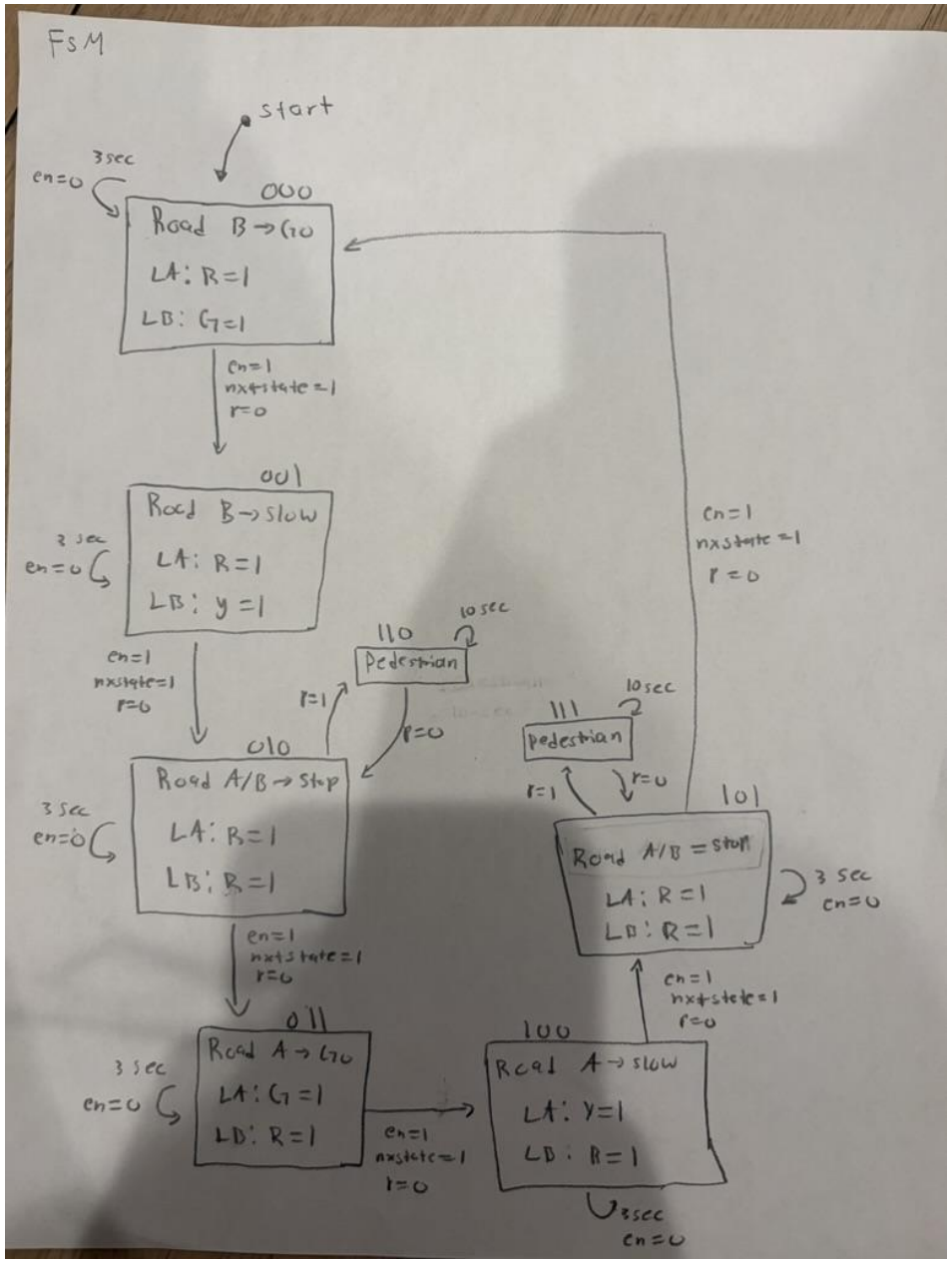
Z = 0 Signal from 40-0 down-counter

Final FSM + Truth Table

Closest Attempt I Got To

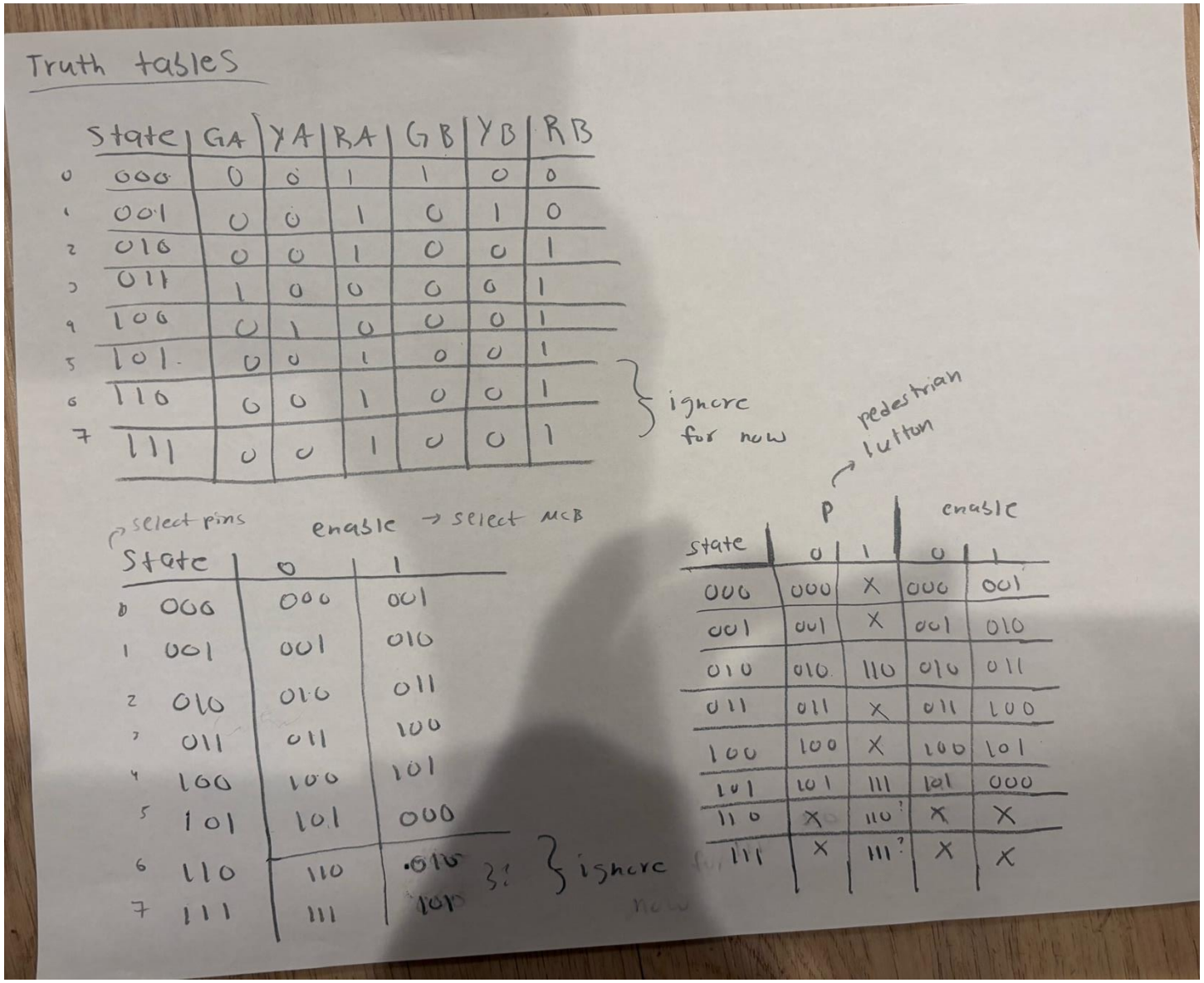
FSM

What I was hoping to implement but wasn't able to due to time constraints



Truth Tables

What I was hoping to implement but wasn't able to due to time constraints

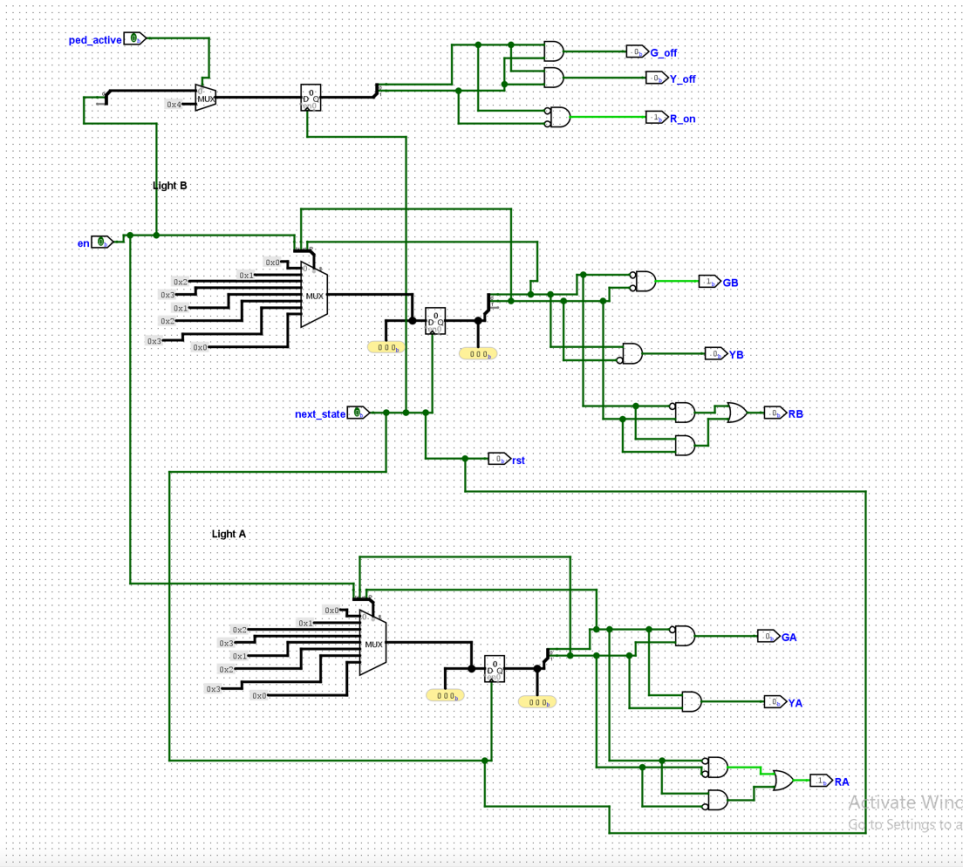


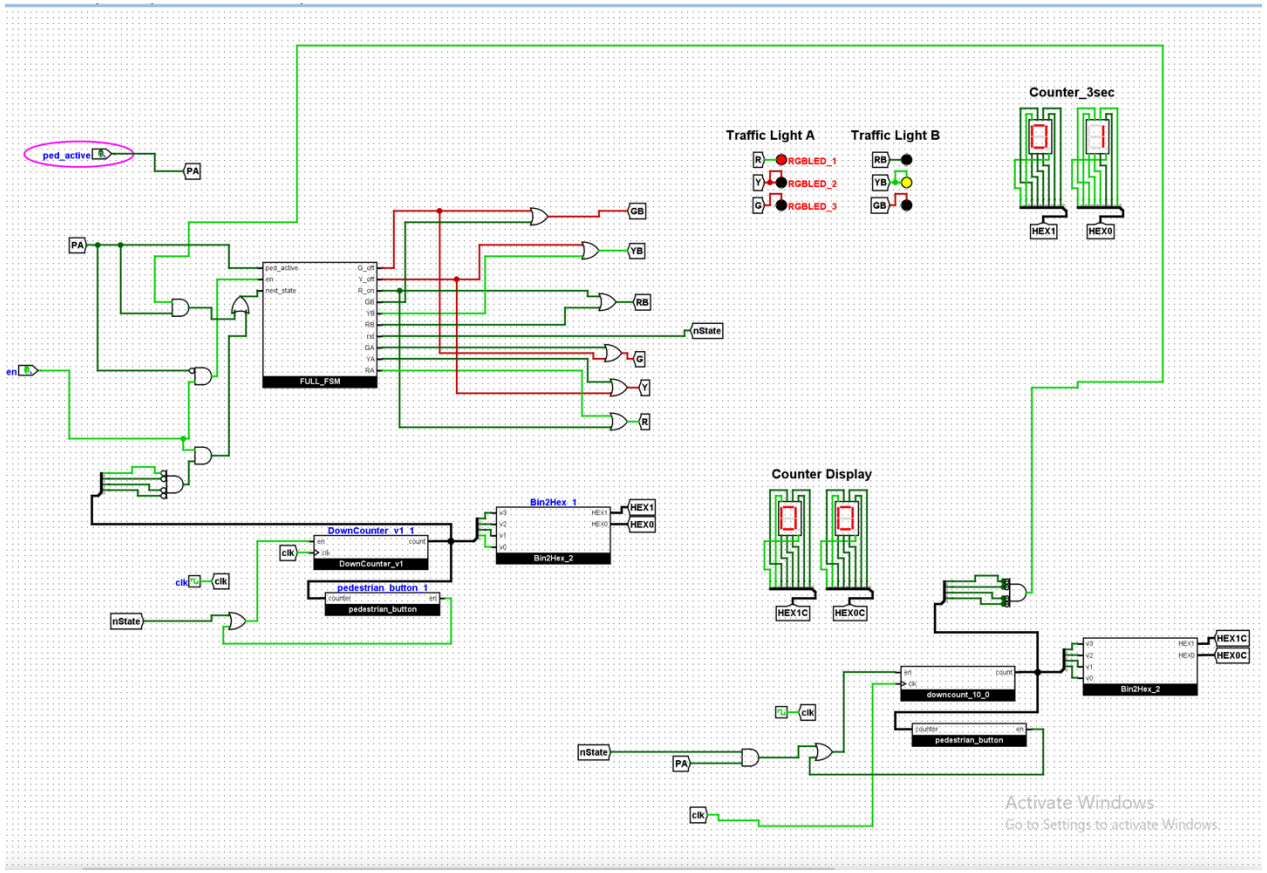
What I wanted to happen:

- The two lights run their normal sequence until a button is pressed. Each light changes every 3 sec.
- When the pedestrian button is pressed, the lights are forced red (or the one that isn’t red finishes its cycle until it hits red). The Red-Red is its own state(s).
- Once both lights are red, they stay red for 10 sec while the pedestrian crosses.
- Once the system detects that the 10 seconds is over, it runs the normal light sequence again…
- Until the pedestrian button is pressed. Cycle repeats.

Final Circuits for Closest Attempt

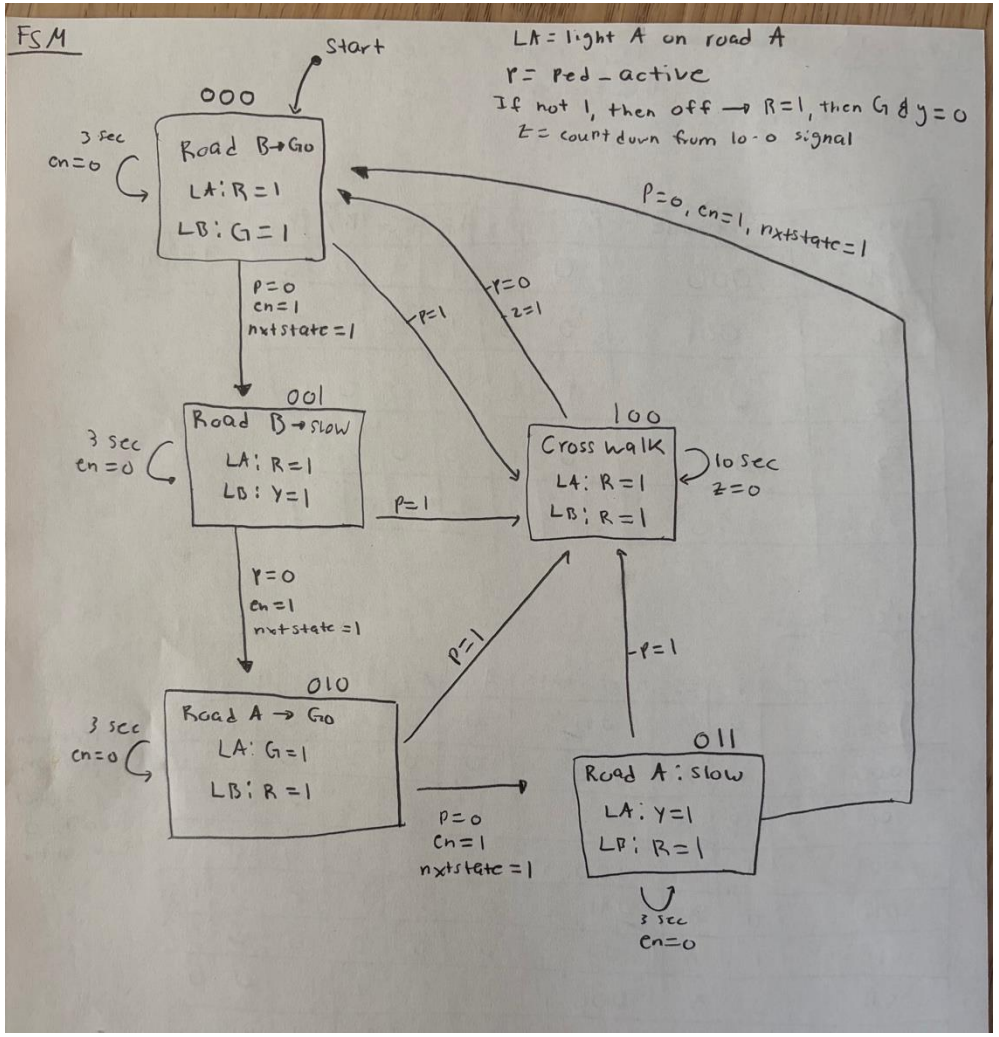
Circuit FSM + Main





FSM that last slide was based on

I wanted to implement what I had in “Final FSM + Truth Table” section, but I wasn’t fully able to due to time constraints with travel, other projects, and going to office hours Monday night with this due Tuesday. But I still wanted to show my thinking. So this is the FSM that I built my closest attempt off of from the slide before this.



All Circuits/Process

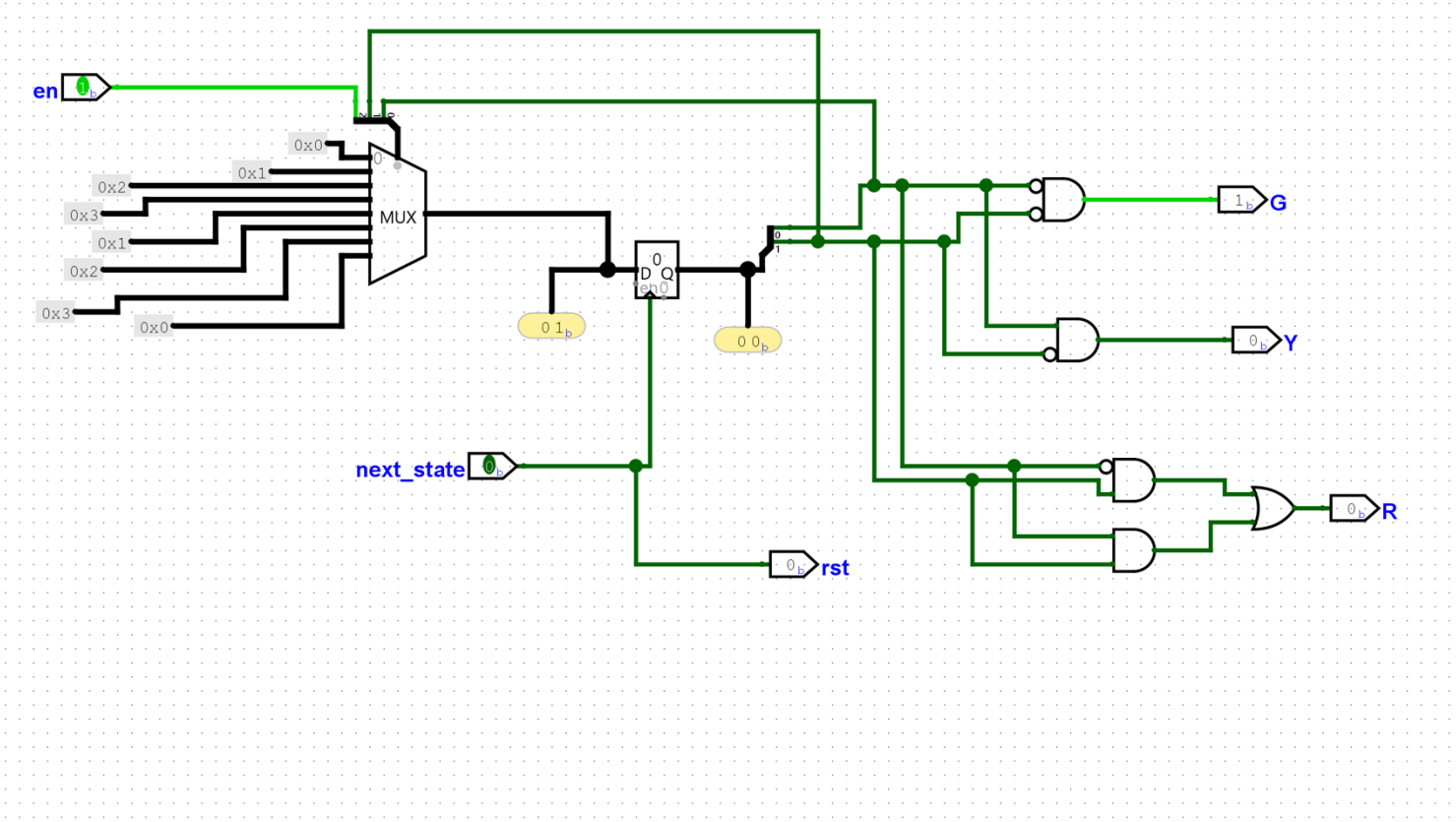
Things I Tried

FSM Circuit Where Light holds for 3 Sec

Trying to come up with circuit or a single light sequence.

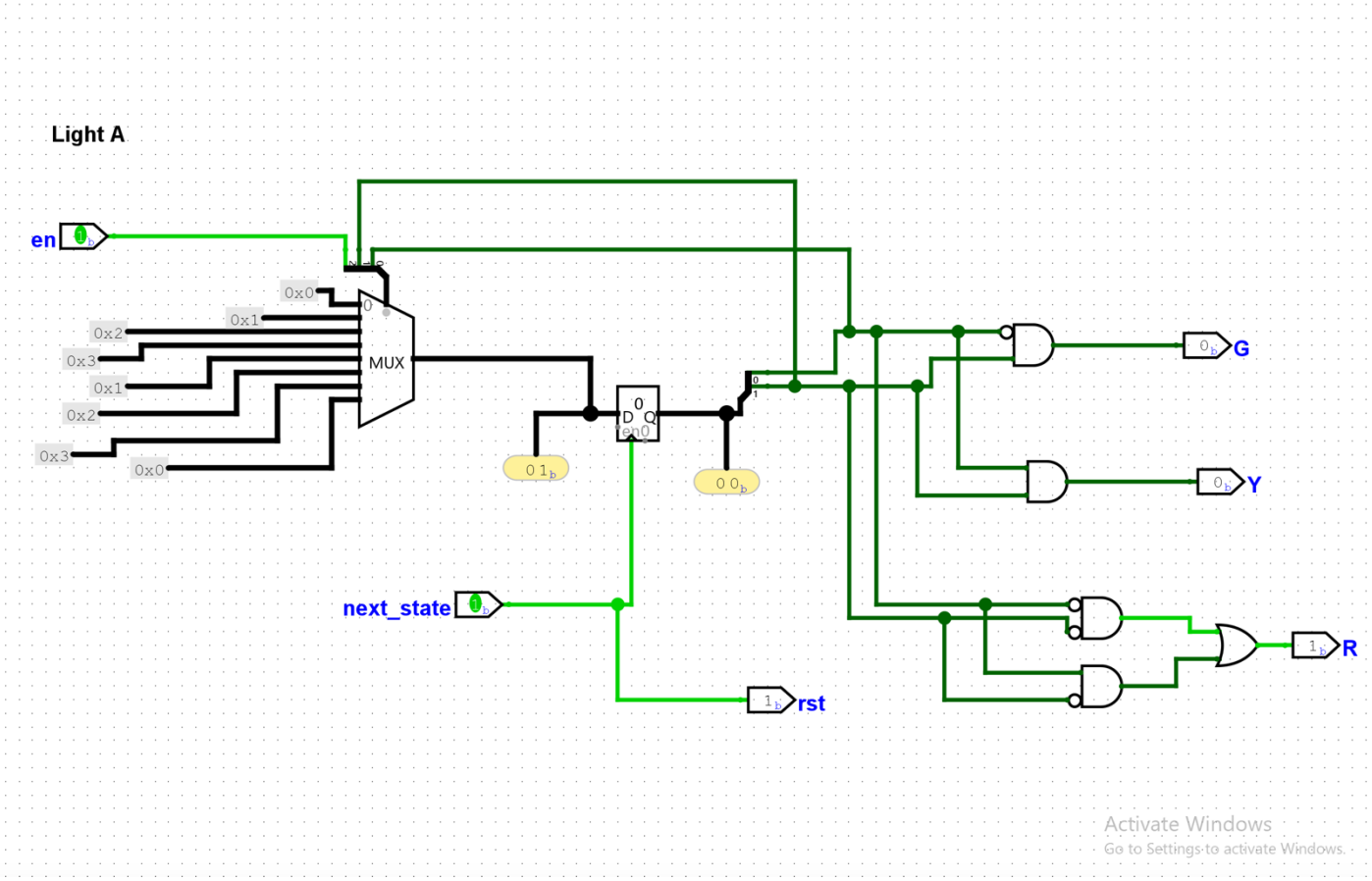
Thinking: Red light is on when R=1. Based off truth table for Light B, Red is on for states 10 and 11. So I can use two AND gates to make R=1 for each condition then use OR gate because R=1 when either one of those states is true.

This was an experiment circuit to see what worked and what didn't

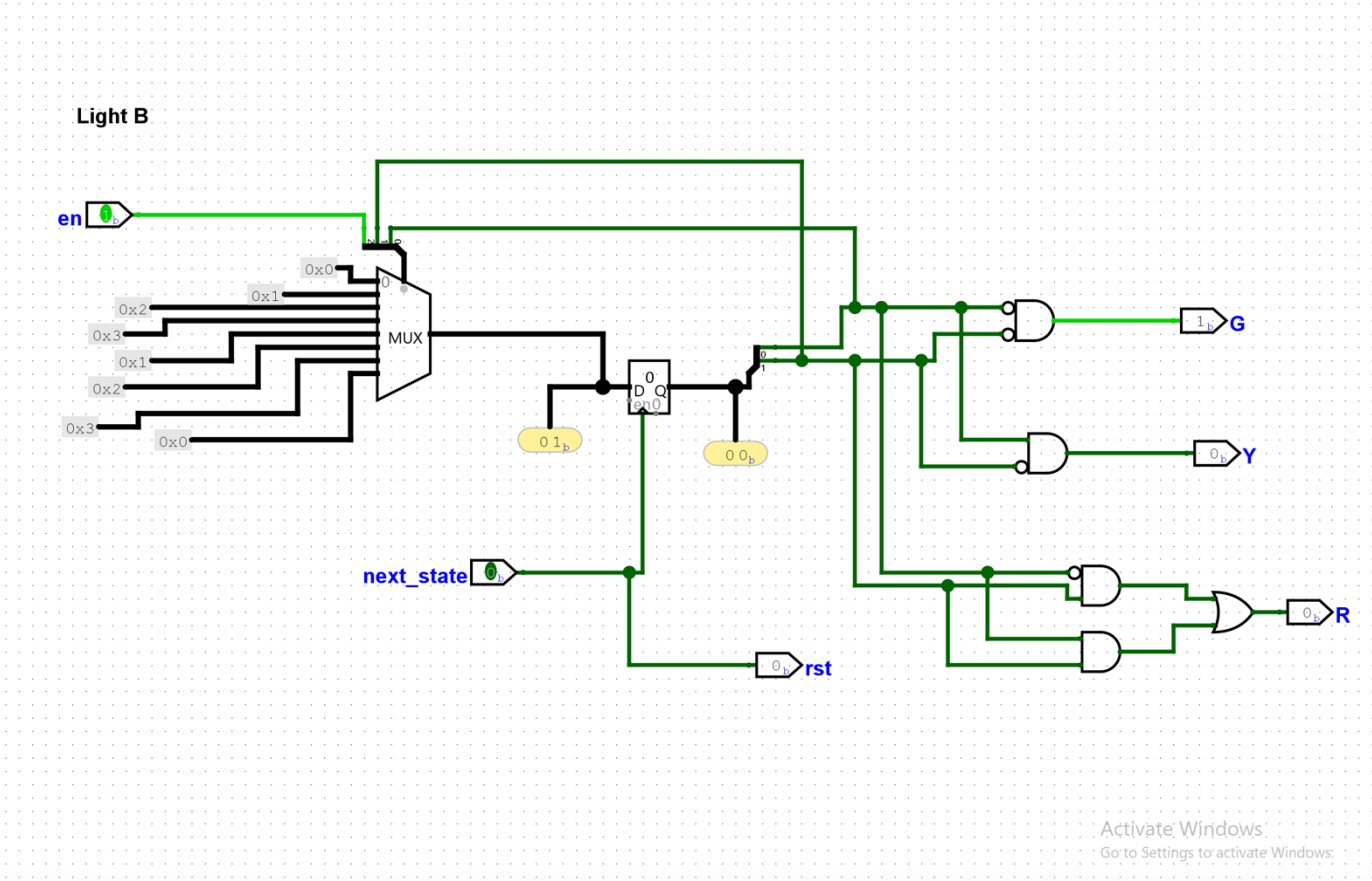


FSM Circuit for Light A

Original thought was to make two separate FSMs for each light (one for each road A and B) and then connect them.



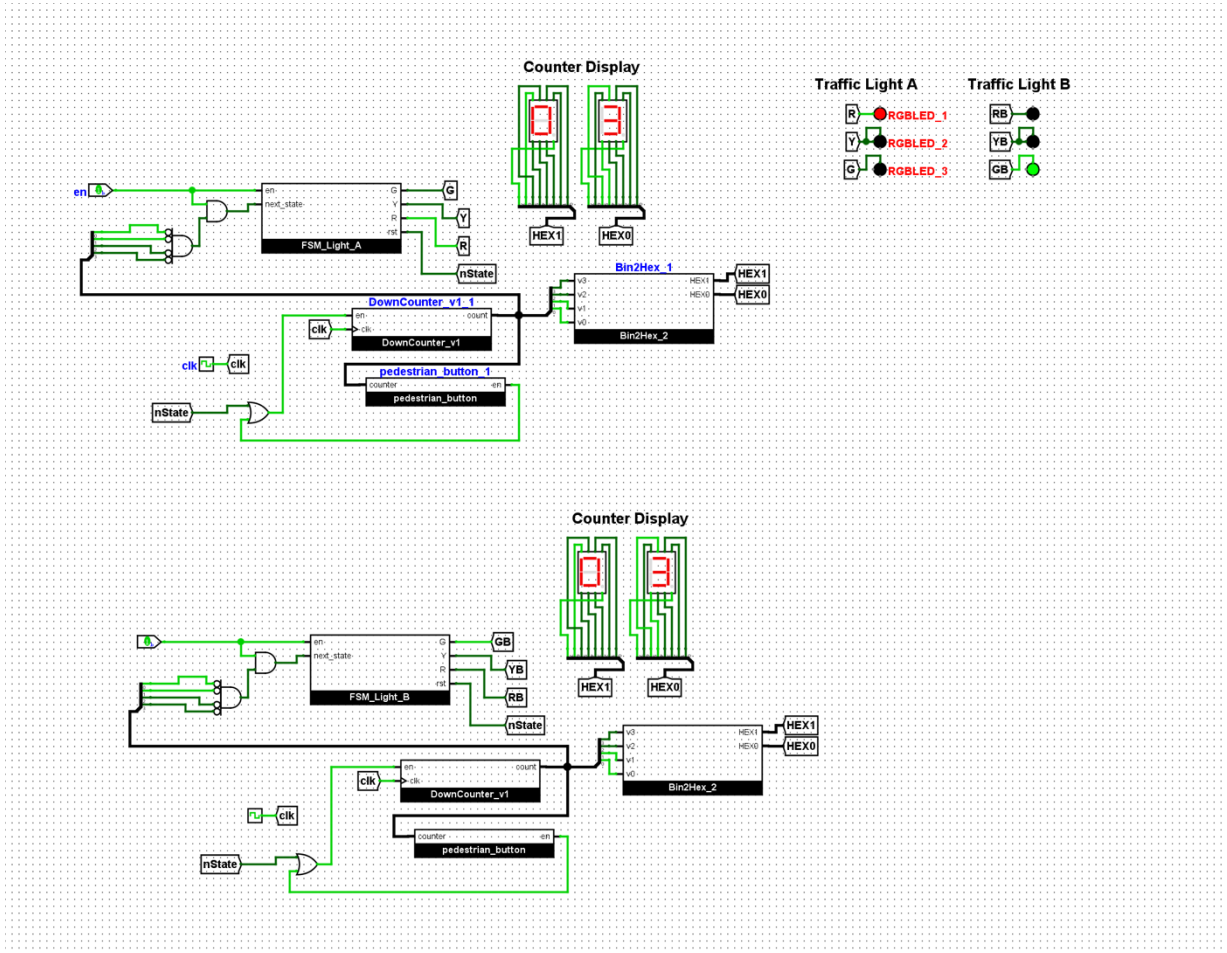
FSM Circuit for Light B



Testing2 Circuit for Light Timing

Trying to implement both the FSMs for the normal light sequence.

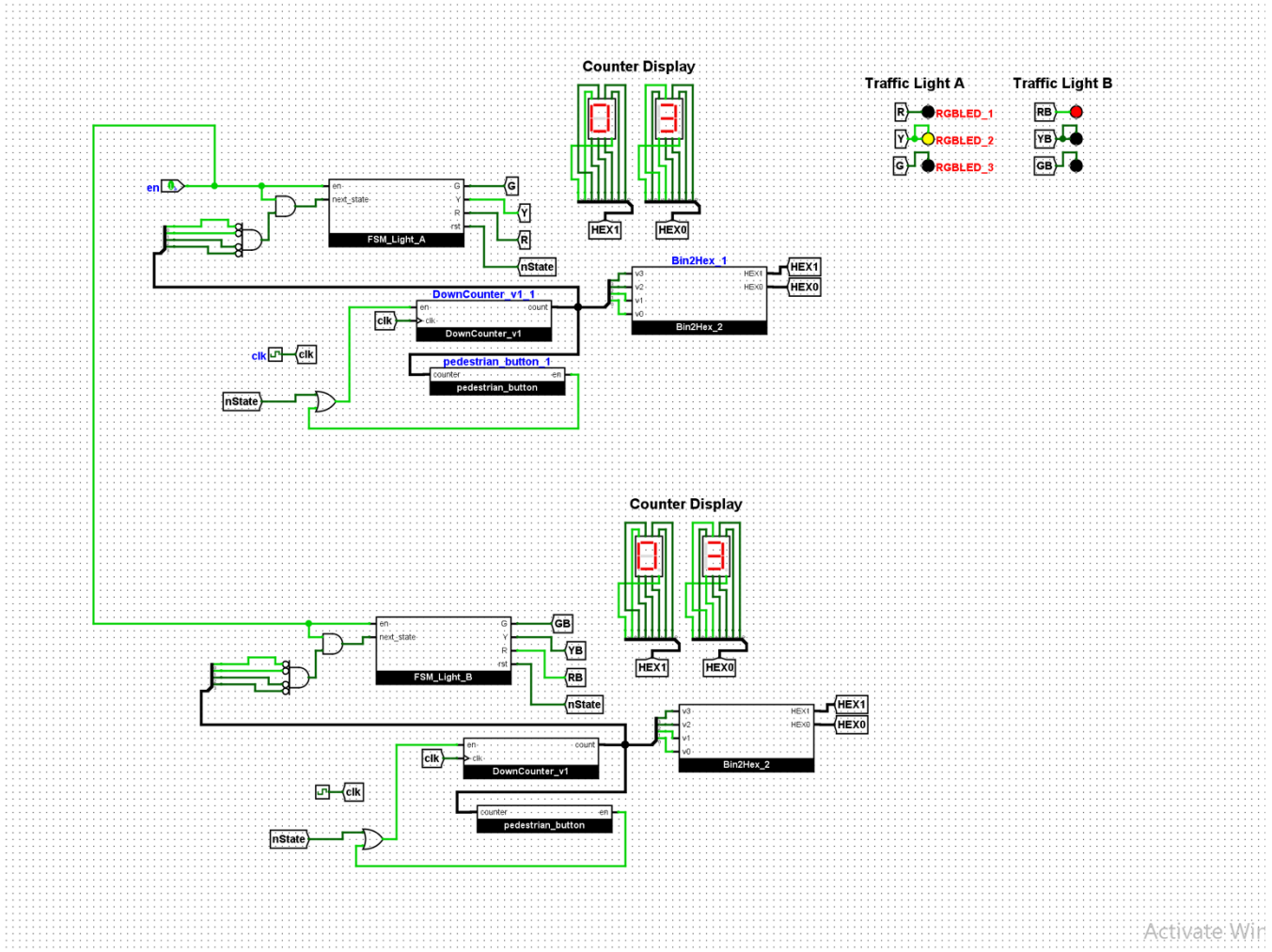
Noticed the nState tunnel needed to be same for both so that they stayed on same timing/switching pattern



Testing3 Circuit

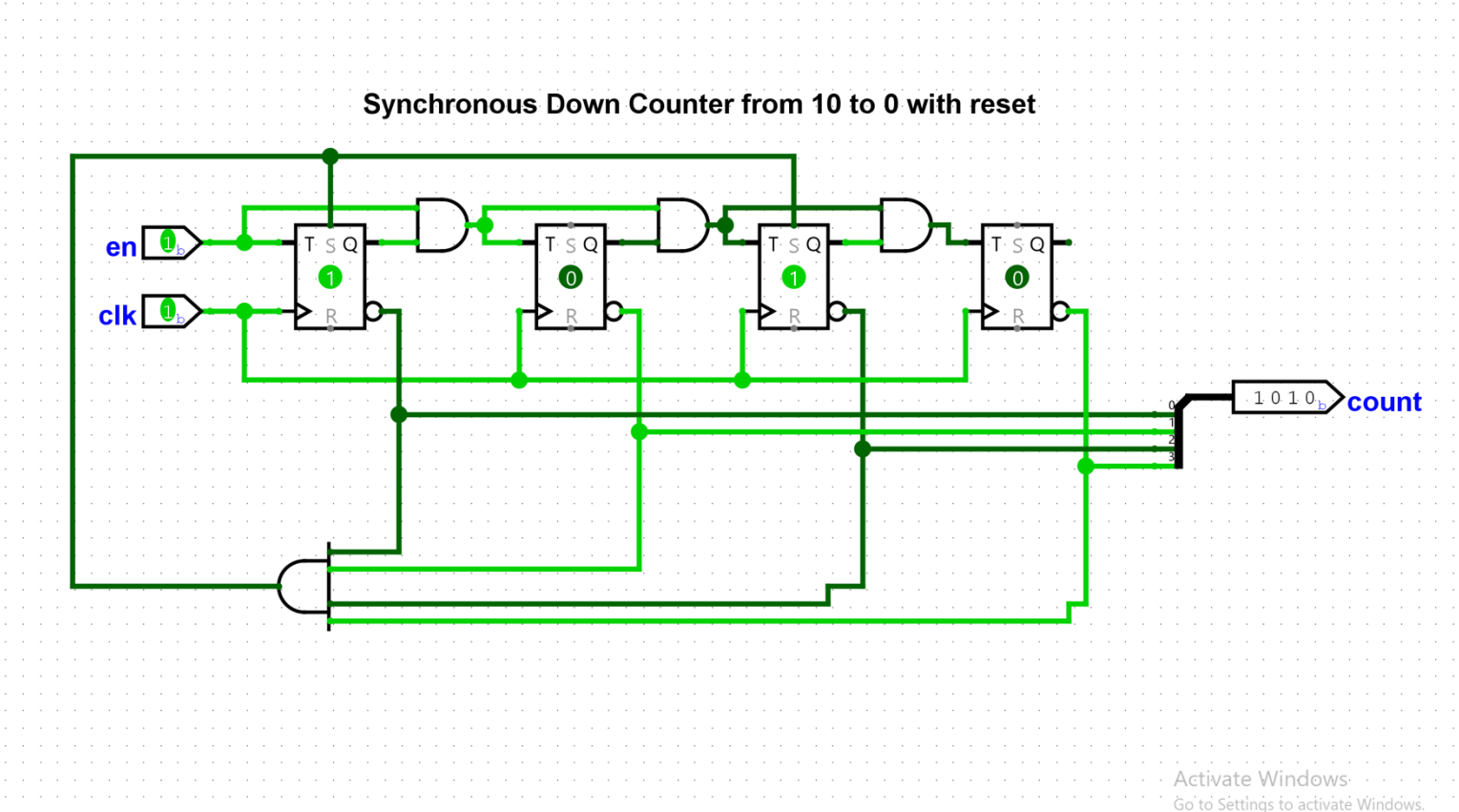
Experimenting with FSMs and the normal light sequence.

Realized I could connect both to one enable button so both lights start at the same time.



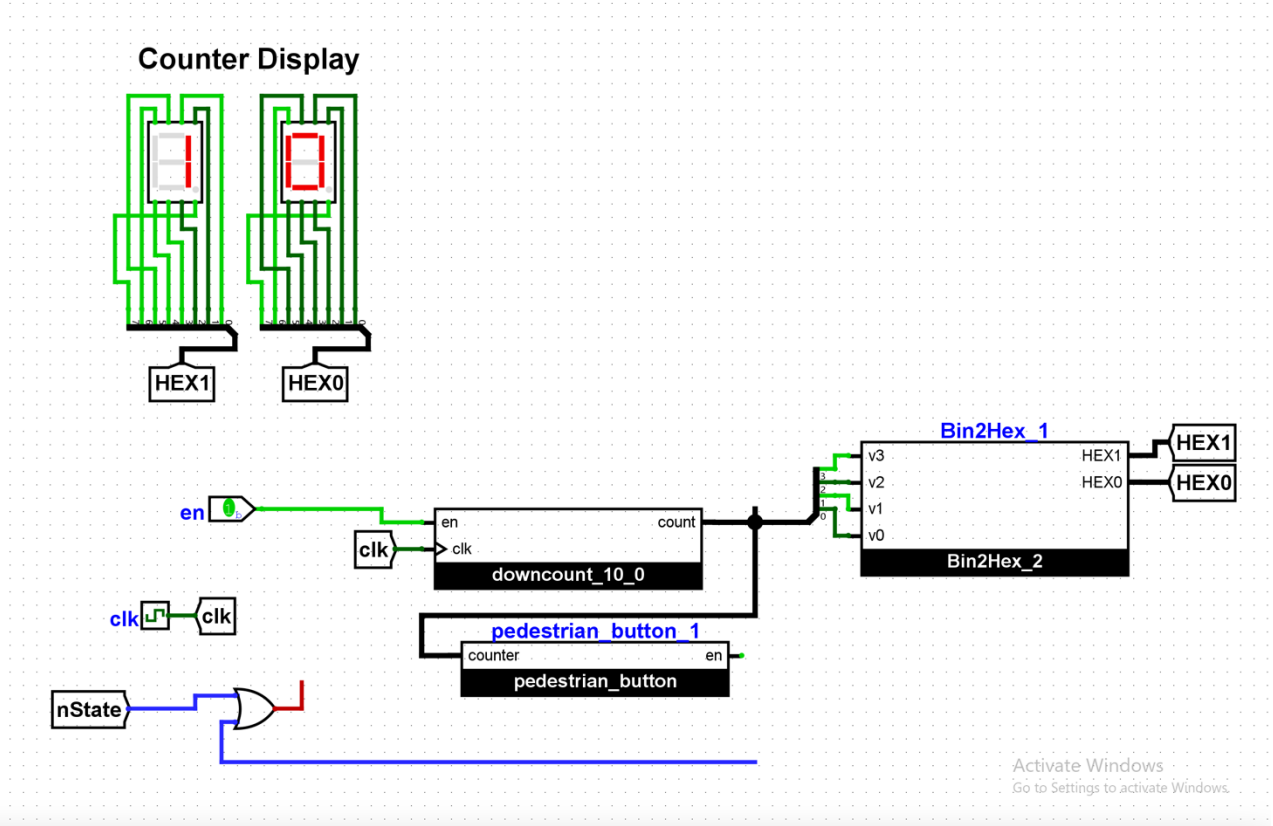
Down-Counter from 10-0

Needed a counter to go from 10-0 for the pedestrian state.



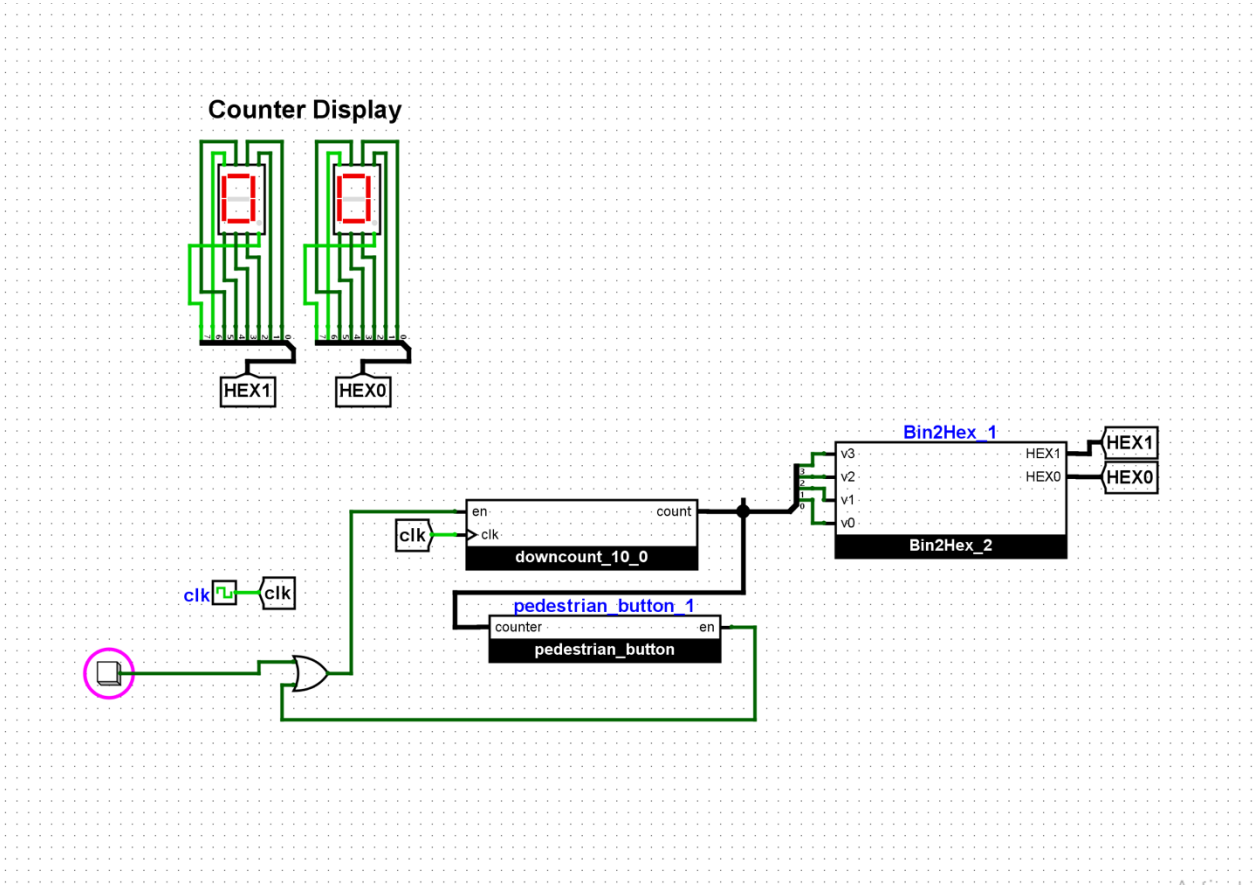
Problem:

I have a circuit that can continuously cycle from 10-0, but how do I get it to only do that when the button is pressed and when both lights are red?



Solution Attempt

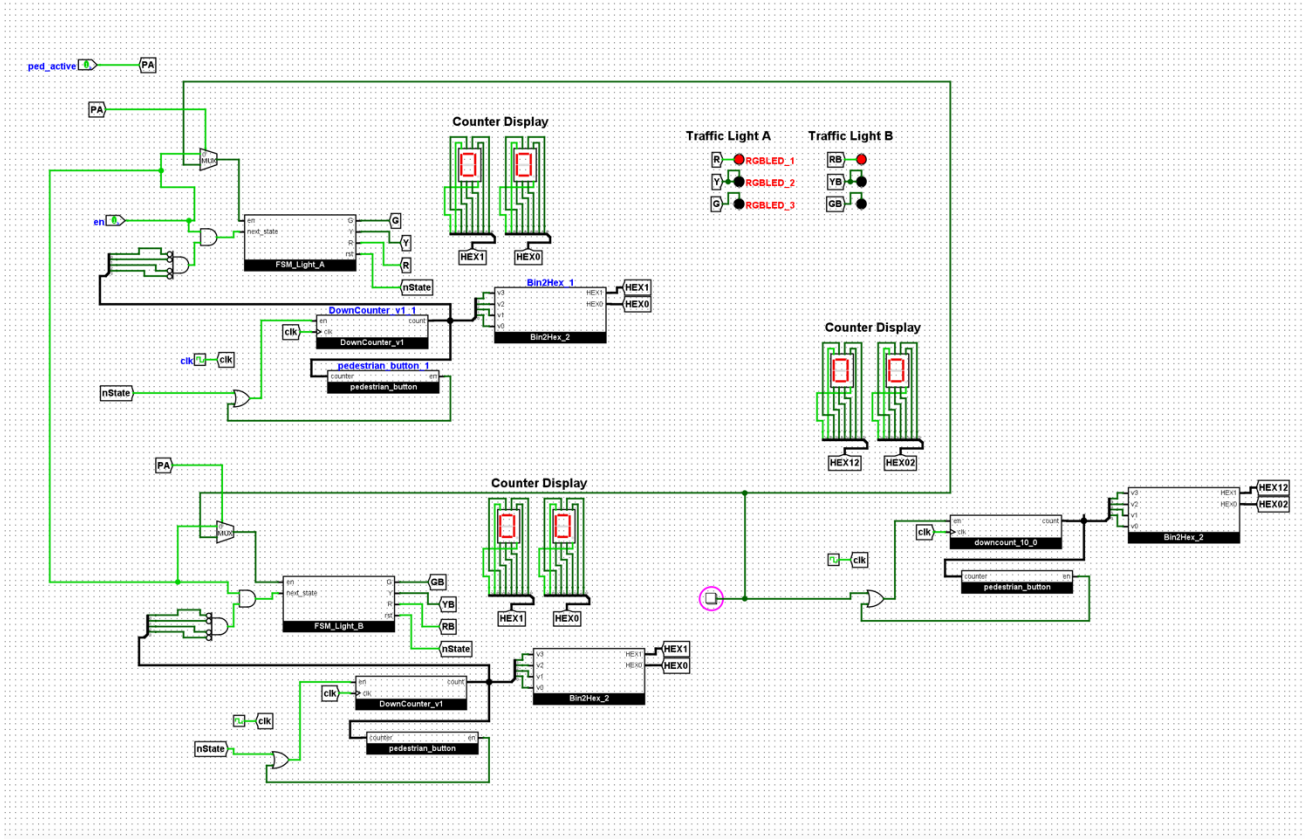
I have a circuit that will count down from 10-0 when the button is pressed and wont have it go from 10-0 until it hits 0 and then button pressed again (ie pushing button when its actively counting down will not make it reset to 10). But how do I connect this to rest of circuit?



Idea 1: Trying to Connect Button to Rest of Circuit

I can use two multiplexers, as shown. When ped_active is 0, it allows normal light flow/loop for the two lights. When ped_active is 1, it freezes the lights and allows me to press button to start 10-0 countdown.

Problem: lights aren't set to freeze at red, countdown doesn't start when both lights red, the rotation of colors between the two lights got all messed up.

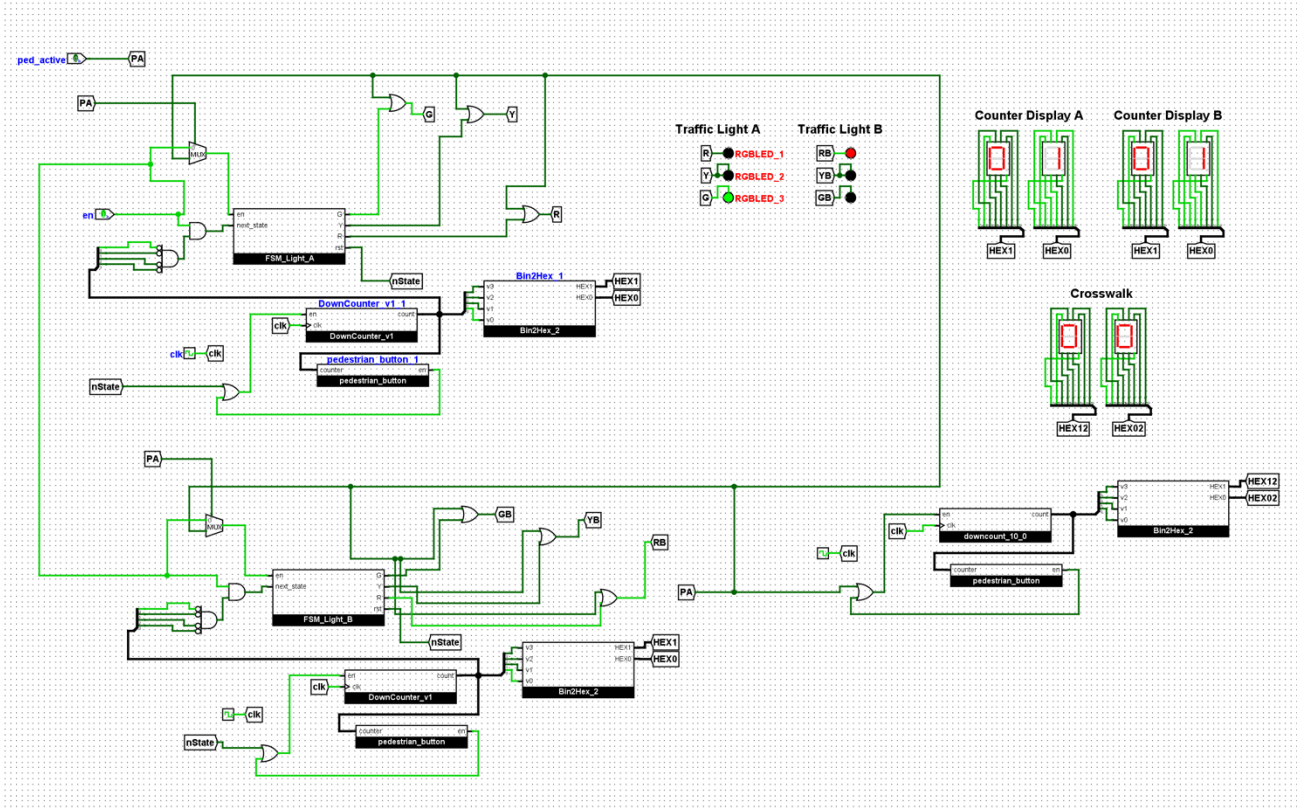


Idea 2: Trying to Connect Button to Rest of Circuit

Thinking: what if I add OR gates to all light tunnels and connect the En to one input and PA to the other input on the MUX.

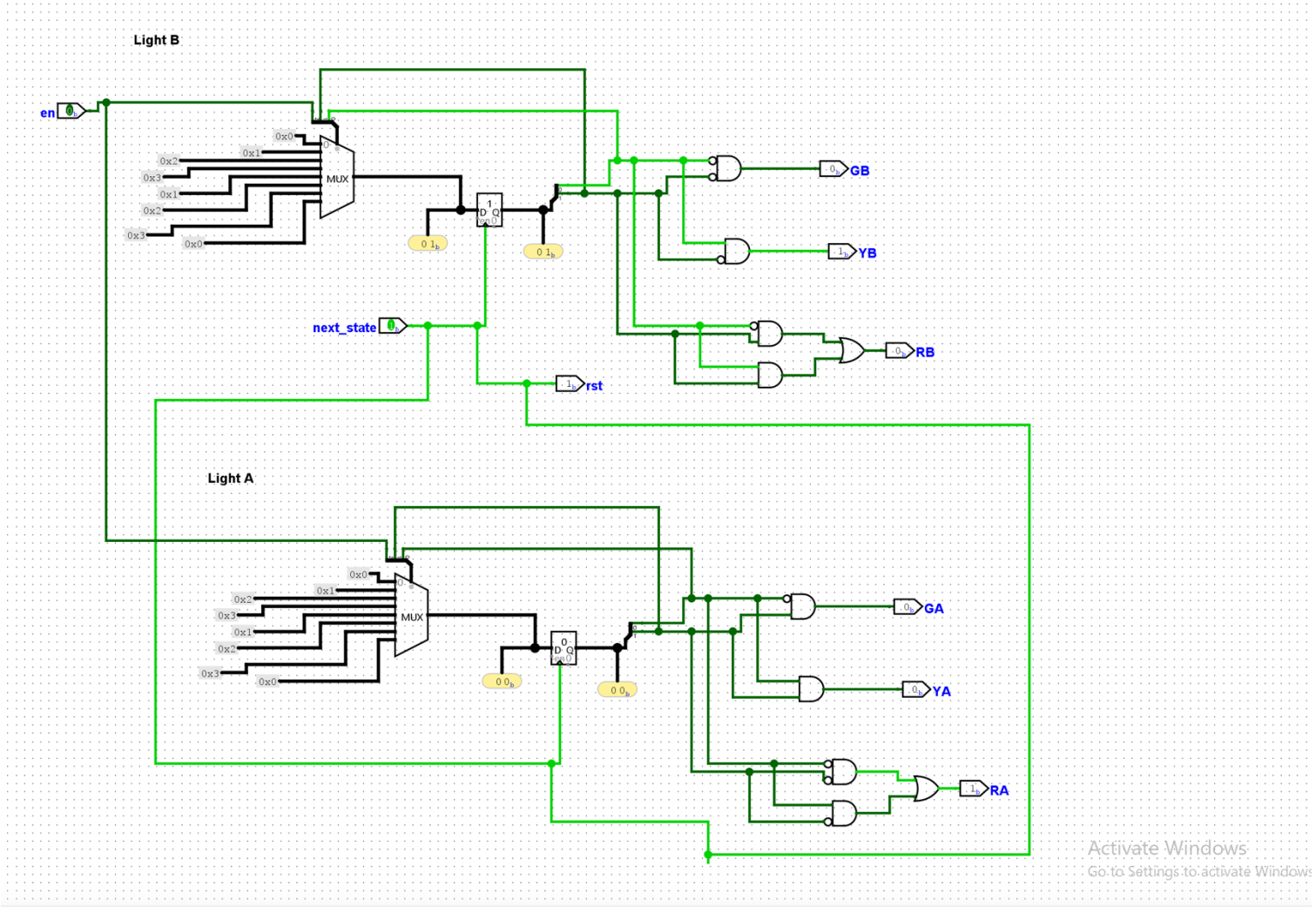
What happened: the lights froze when ped_active was 1 and the 10-0 counter started. When ped_active became 0, the lights returned to their normal cycle.

Problem: all lights turn on and freeze. When ped_active is 1, the counter loops 10-0 instead of stopping at zero.



Global FSM for Lights

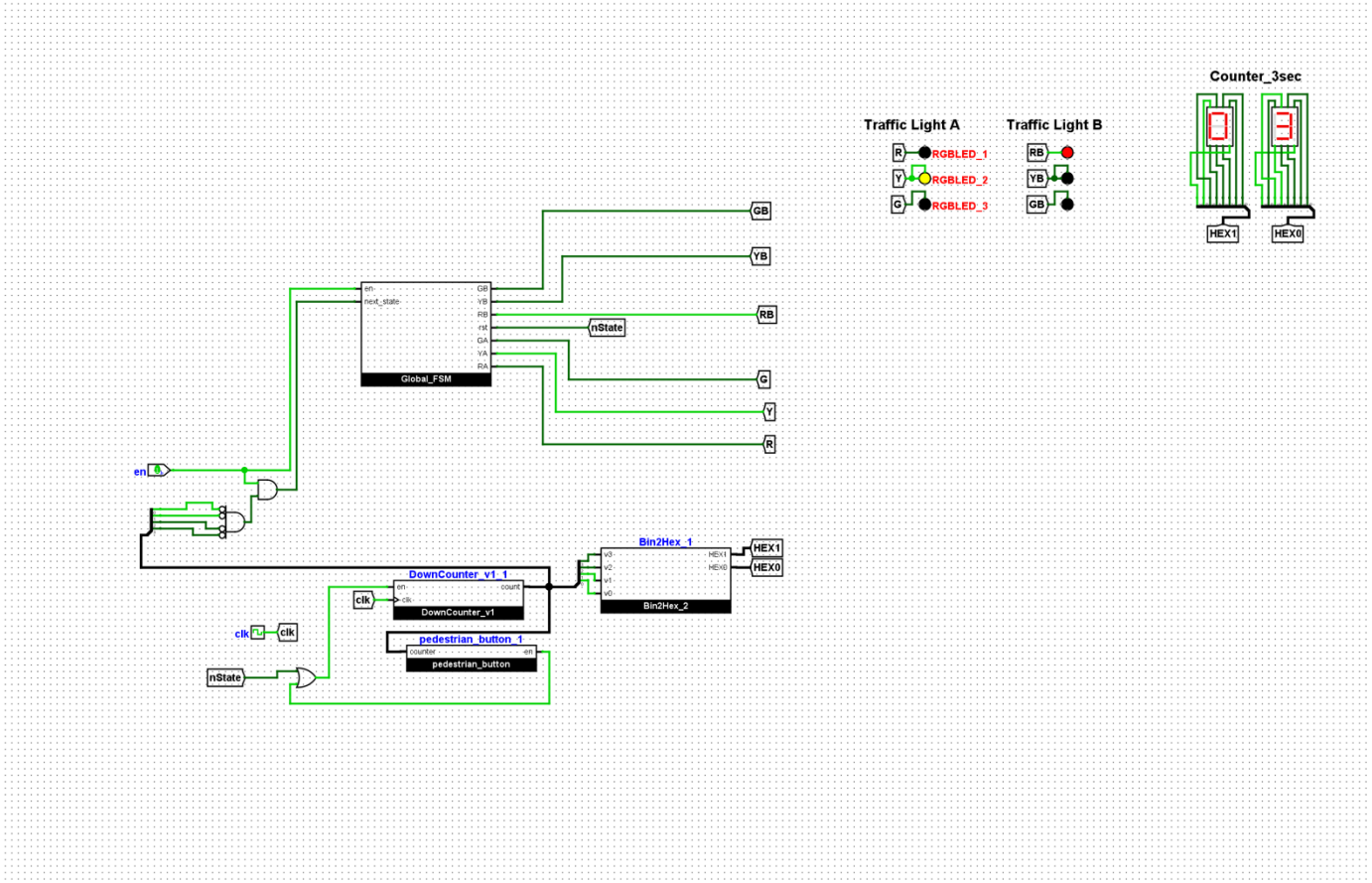
Both lights in one circuit FSM and under one enable. Helping clean up the circuitry in the main.



Activate Windows
Go to Settings to activate Windows.

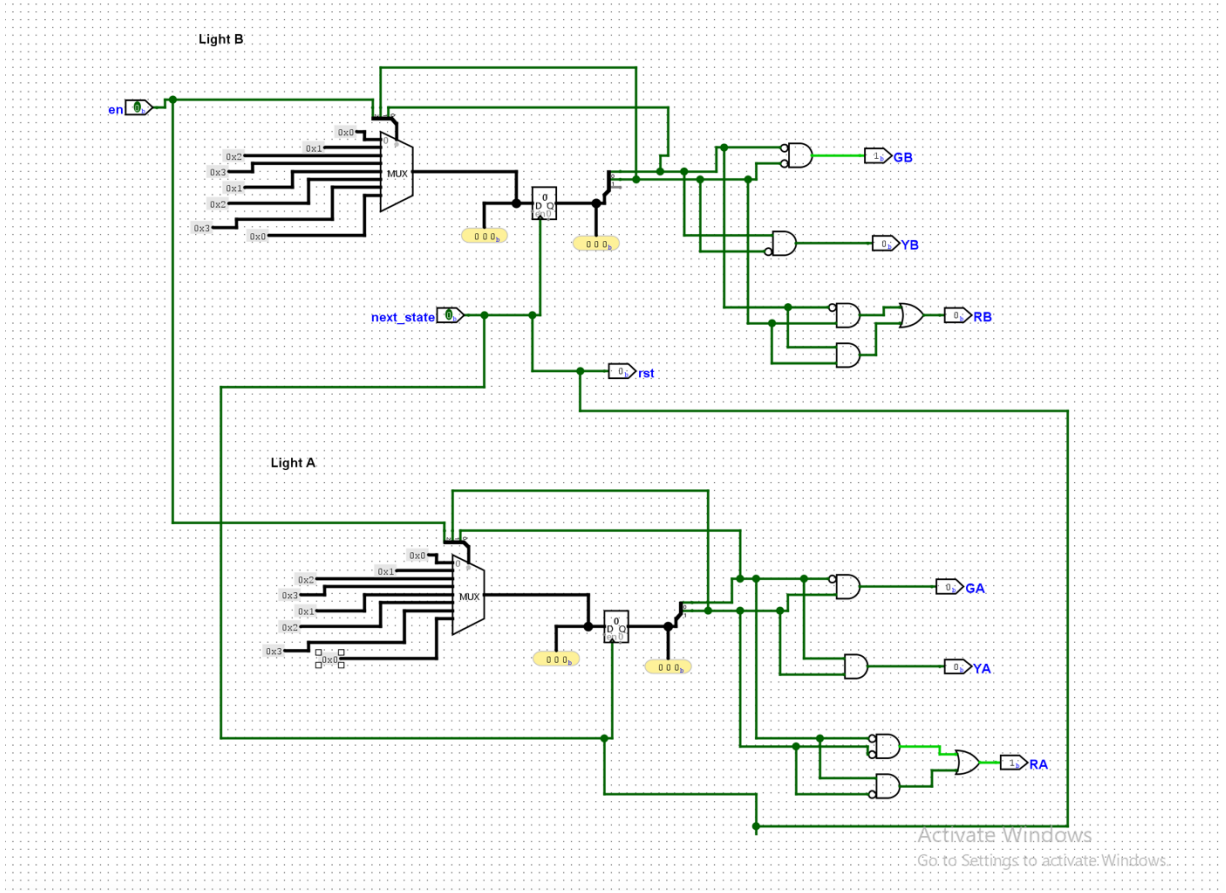
Global FSM implemented for Lights

Work getting too chaotic, so needed to make it more digestible in the main circuit.



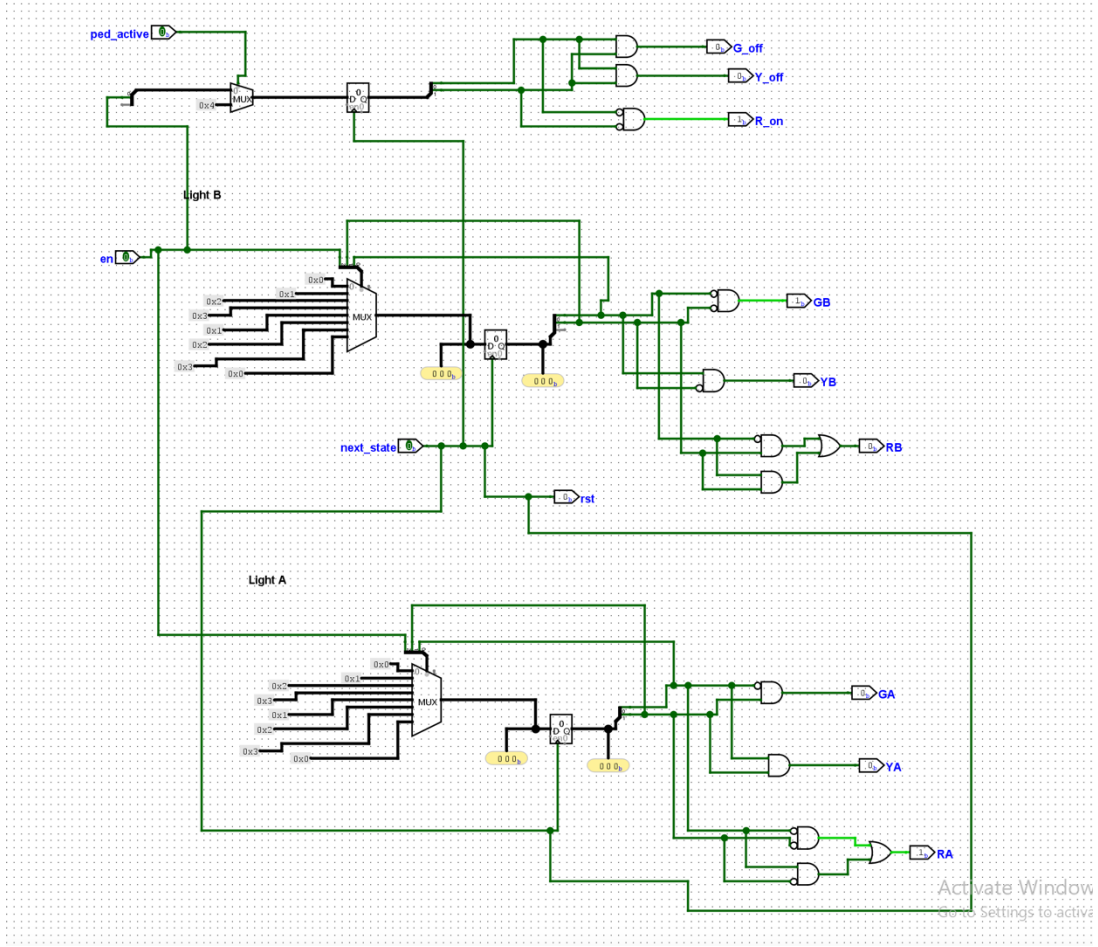
Global FSM Using 3-bits for Lights

Realized if have 5+ states, I can’t have everything in 2-bits since state 4 is 100 in binary. So I converted everything to be 3 bit



Trying to Connect Pedestrian Button/Logic to 3-Bit FSM for Lights

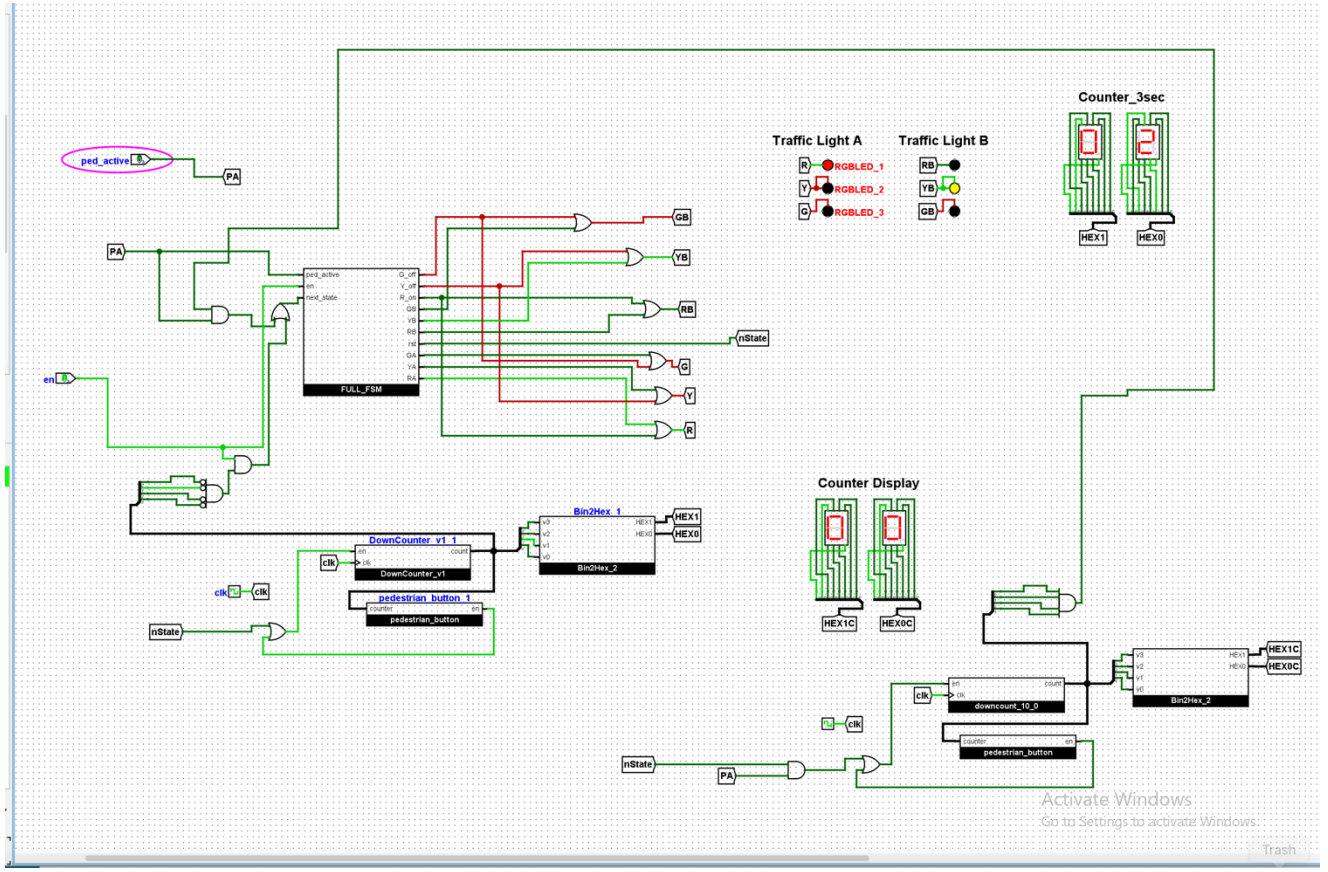
Thinking: When ped_active is on 1, I want to go to state 4 where we turn green and yellow off while keeping red on. But when ped_active is 0, I want to run the lights normal. So I'll try to use a MUX.



Idea 3: Trying to Connect Pedestrian Button to Rest of Circuit

What happened: When En is 1 and ped_active is 0, the lights run the normal sequence. When ped_active is 1, the red lights hold/stay and the 10-sec countdown begins. The circuit resumes normal light sequence when ped_active is 0.

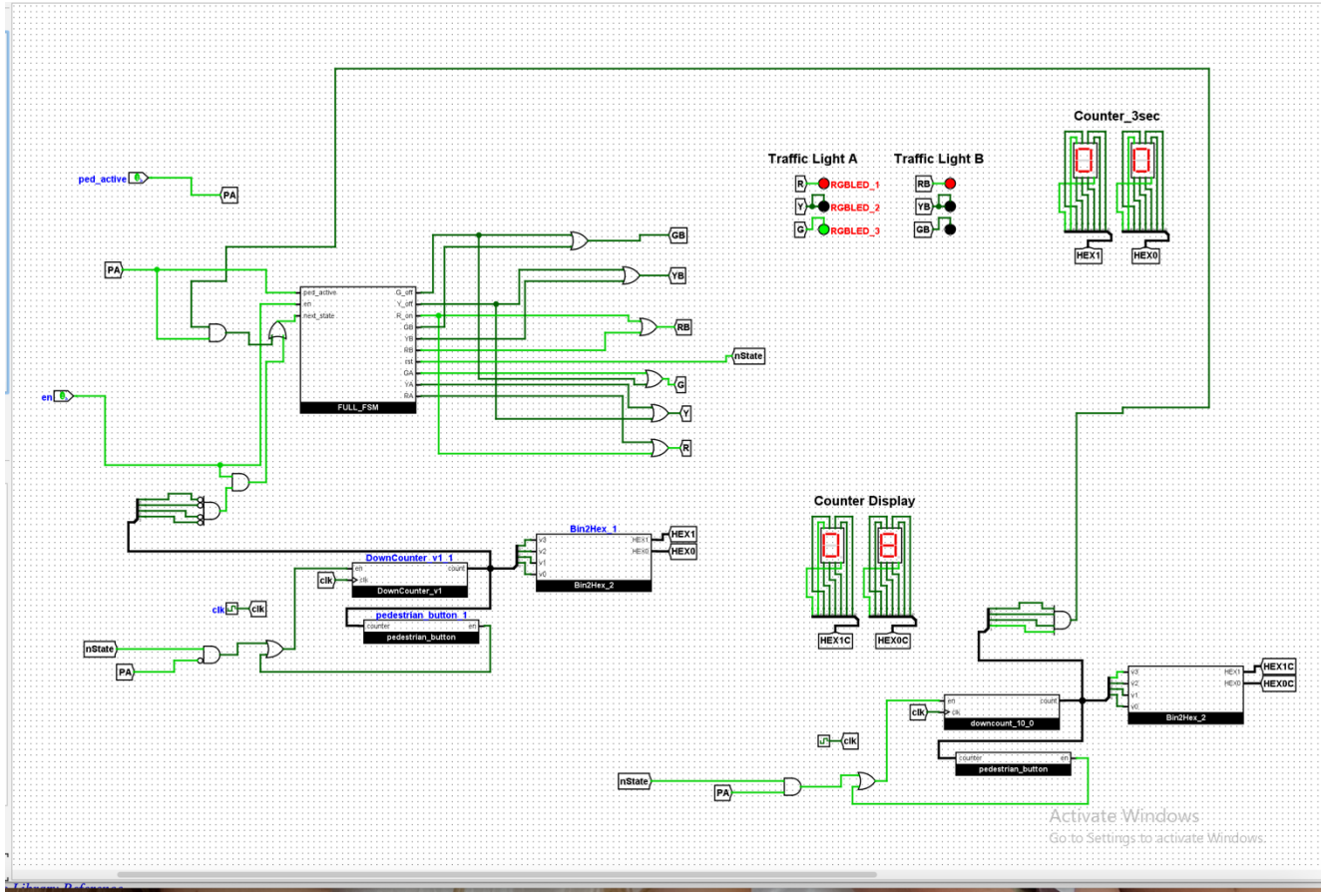
Problem: the red lights stay but the green and yellow still try to cycle through their sequence when ped_active is 1. The 10-sec countdown keeps cycling until ped_active is 0 rather than stopping after 1 cycle.



Idea 4: Trying to Connect Button to Rest of Circuit

What happened: Tried messing with logic gates. When En is 1 and ped_active is 0, the lights run the normal sequence. When ped_active is 1, the red lights hold/stay and the 10-sec countdown begins. The circuit resumes normal light sequence when ped_active is 0.

Problem: the red lights stay but the green or yellow still freeze with red when ped_active is 1. The 10-sec countdown keeps cycling until ped_active is 0 rather than stopping after 1 cycle.

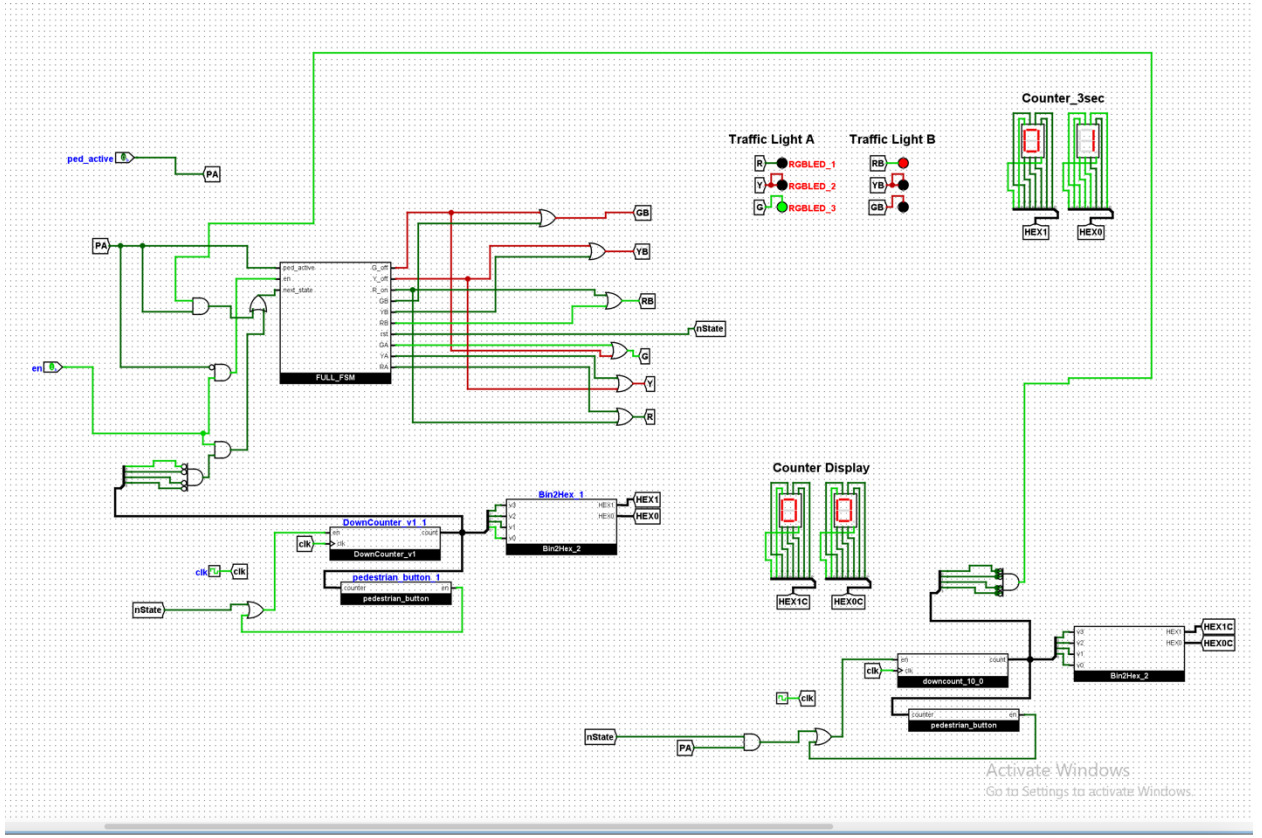


Idea 5: Trying to Connect Button to Rest of Circuit

Thinking: I only want to enable the light sequence when ped_active is 0 and enable is 1. So why not use an AND gate so that when ped_active is 1, the light sequence "turns off."

What happened: When ped_active is 1, the red lights stay. When ped_active is 0, the normal light sequence continues.

Problem: the moment ped_active becomes 1, the red lights cue but whatever light was on in normal sequence also freezes. Still can't get 10-sec timer to stop after 1 cycle.

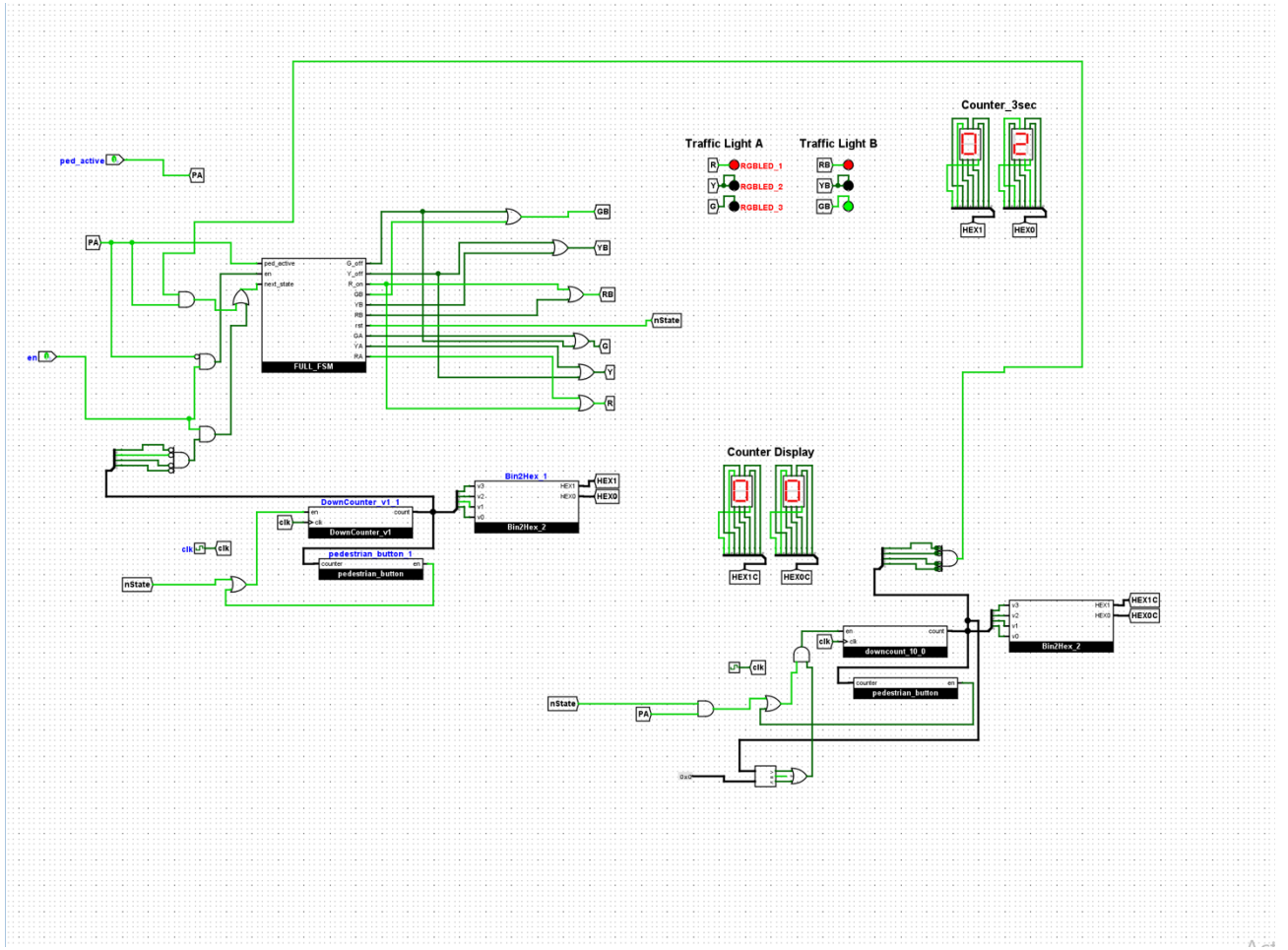


Idea 6: Trying to stop timer after 10-sec

Thinking: Use a comparator circuit so that when it detects that the timer hits 0, it turns off the counter enable and/or the ped_active state.

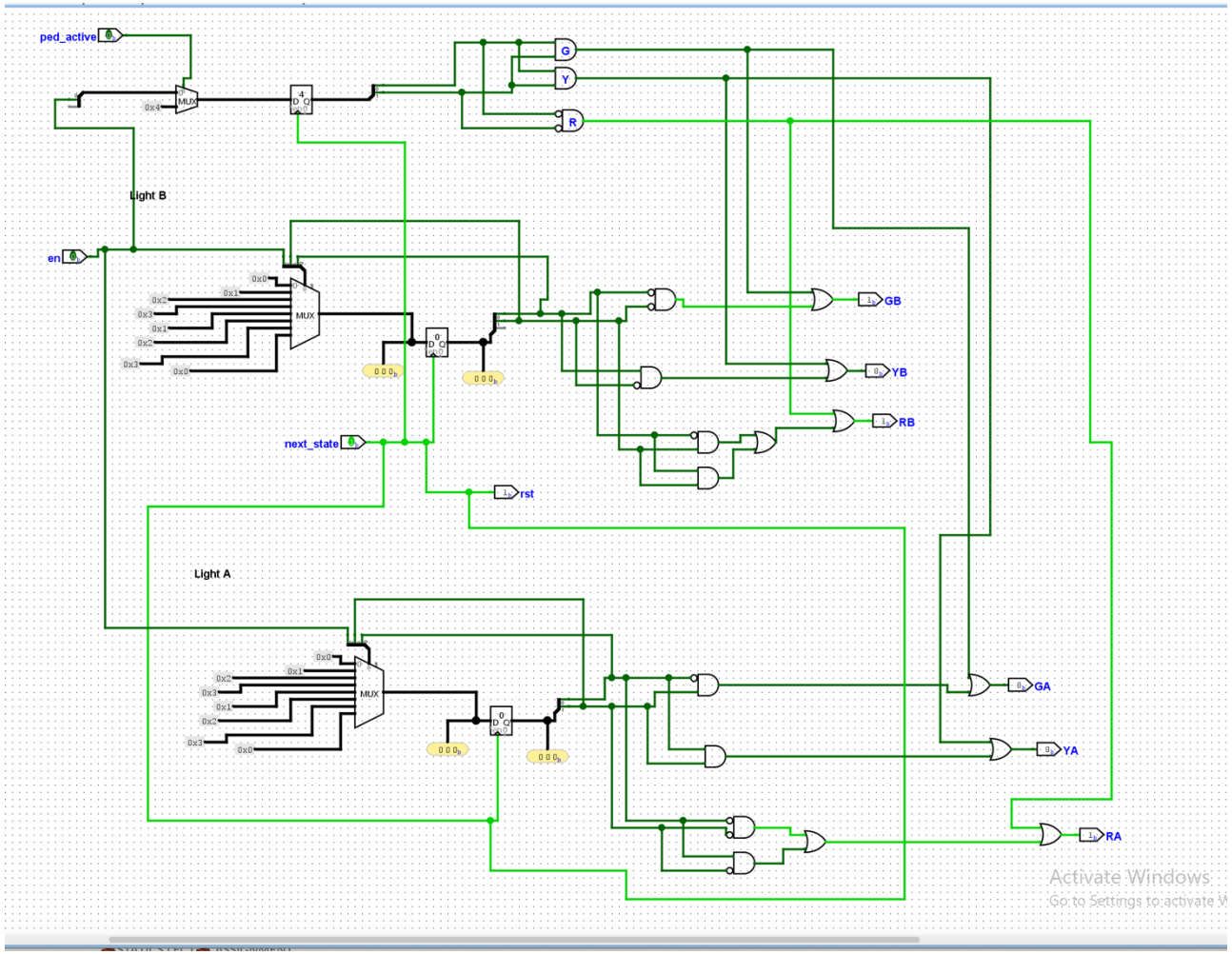
What happened: When ped_active is 1, the 10-sec countdown began and stopped at 0.

Problem: the moment the counter became 0, the counter stopped. However, I lost the ability to reset it. I had to manually reset it by connecting and disconnecting the = output to the OR gate. Also stopped the lights from staying red since it turned off the next state for the pedestrian circuit.



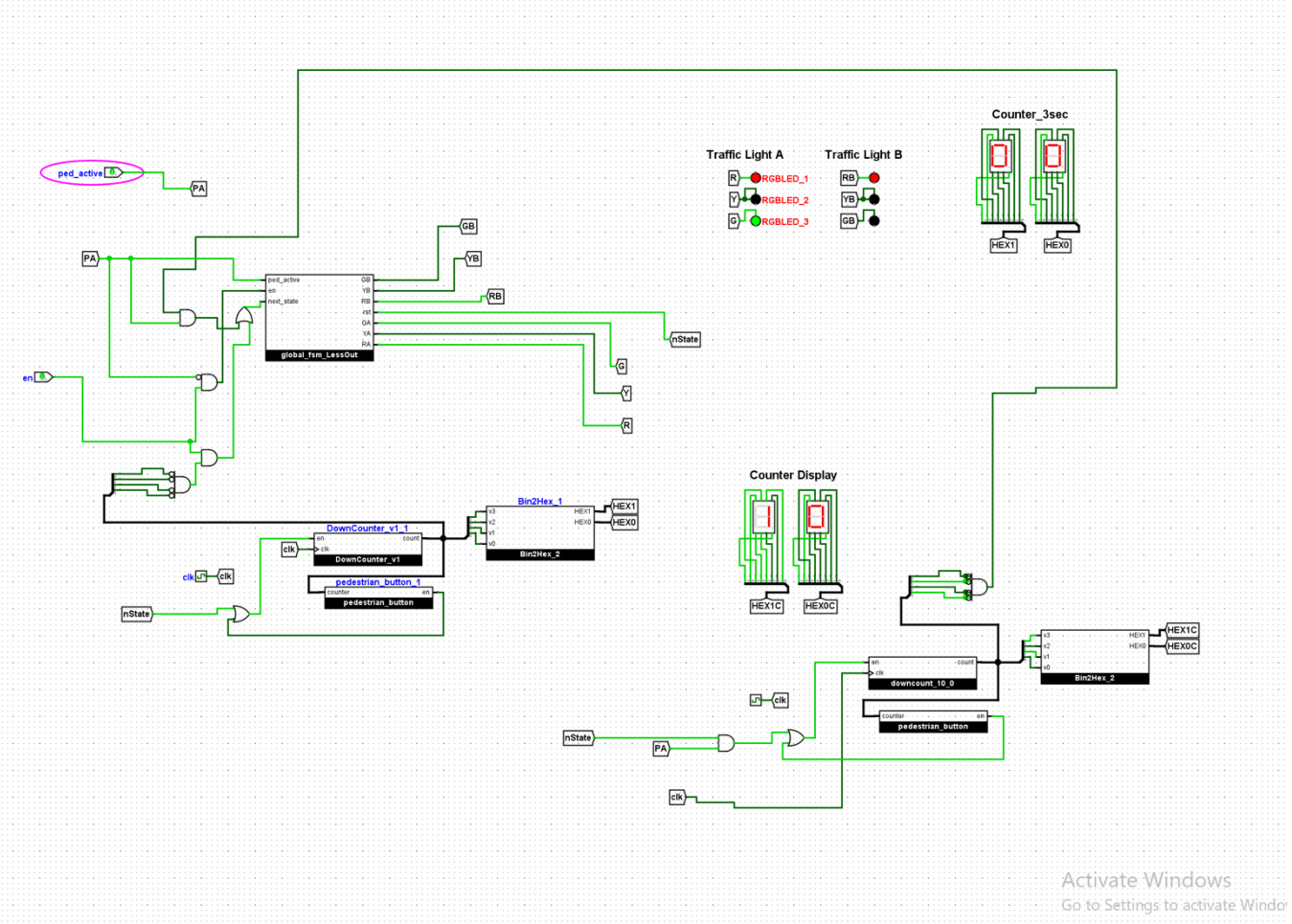
Idea 7: Global FSM with Less Outputs

Thinking: Rather than having separate outputs for state 4 (all-red), why don't I just connect that logic to the outputs I already have for the two traffic lights with OR gates.

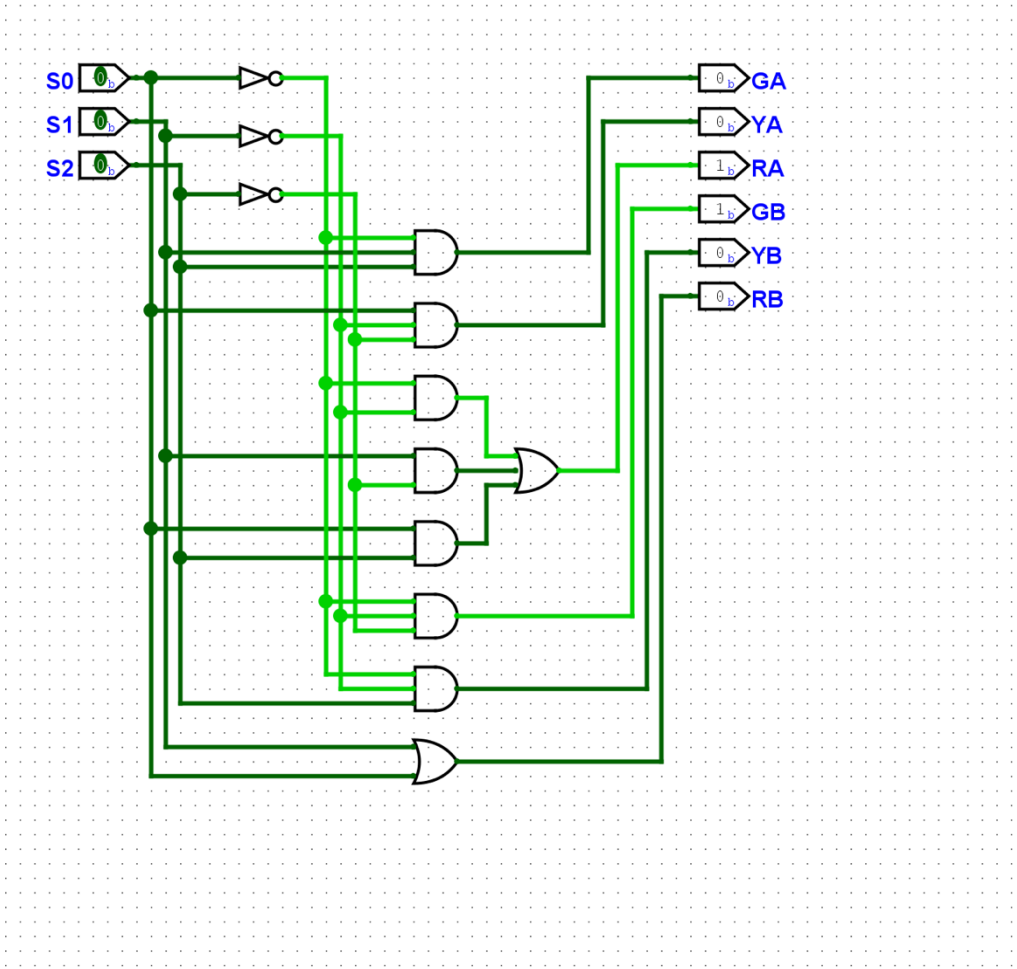


Idea 8: Implementing FSM from Idea 7

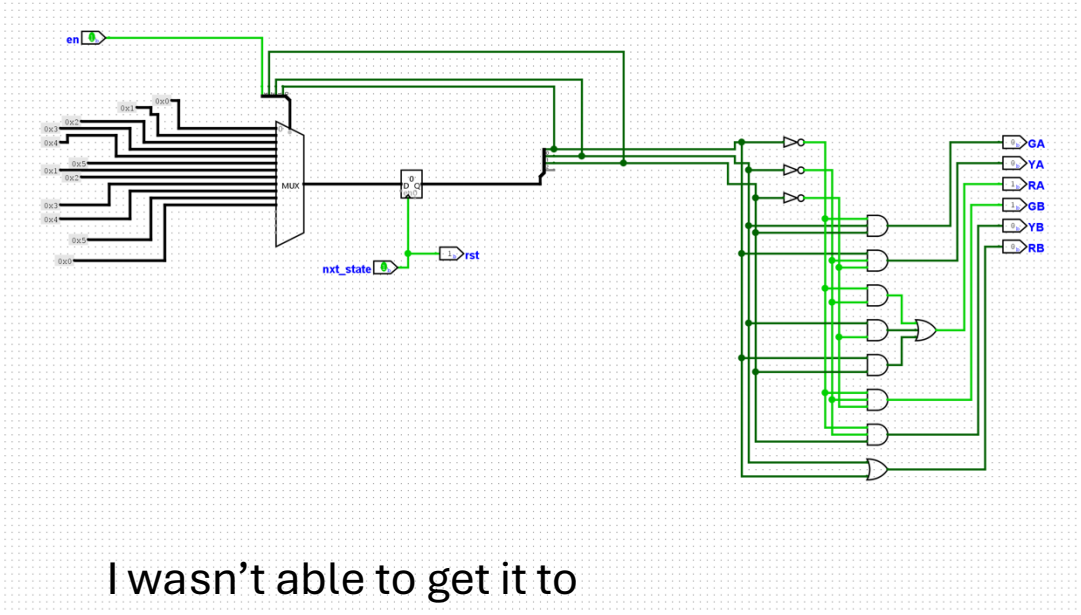
Problem: Still end up with normal light sequence running when ped_active is 0. But when ped_active is 1, both stay red and the 10-0 countdown starts, however, the lights also freeze at whatever state when the button is pressed (ie green here for light A).

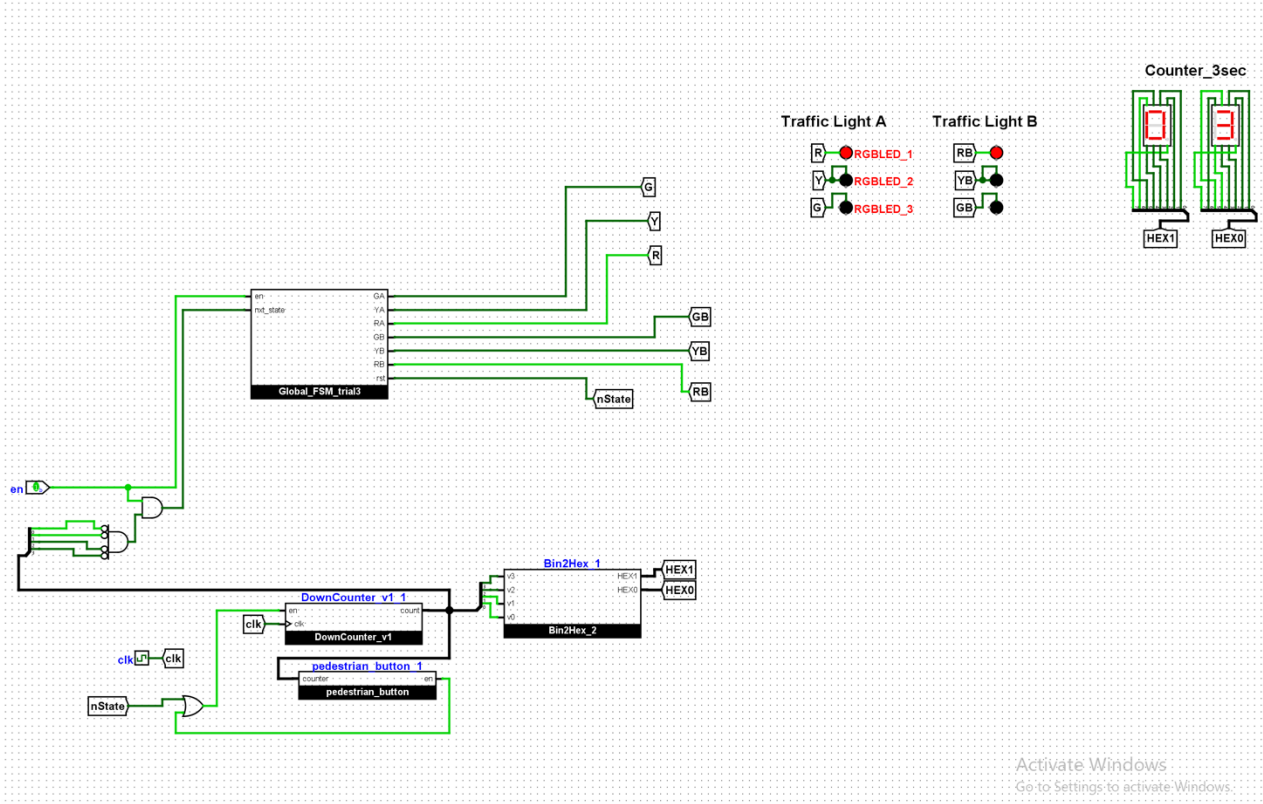


Realized: Need a state(s) where both lights are red. Circuit for light outputs built via truth table



Tried to implement logic into FSM





I wasn't able to get it to attach right to run through the correct FSM light sequence (without the button--states 000 to 101)